

Aftershock

Building Safety Coordinator • Kestrel Bay

15 DAYS • 61 STOPS • GRADE 12 • AFTERSHOCK

THE OPENING CARD

A magnitude 6.8 earthquake hit Kestrel Bay three days ago and the town is still standing, unevenly. You are the building safety coordinator, which means you decide which buildings people are allowed back into: green to enter, yellow for restricted use, red for nobody. Four hundred households are sleeping in halls waiting on those decisions, the hospital has a yellow placard on its front door, and the aftershocks have not finished. Upper Town sits on granite and lost its chimneys. The Flats sit on eighty-year-old fill and lost their streets.

Two towns, one earthquake

BEFORE THE DAY — GREET

Two agencies, one office, and nobody has been introduced

The office is half council and half seconded engineers who arrived on Saturday, and by Friday somebody will have to take a decision on somebody else's assessment. Get round the building and put a name to as many of them as you can now, because a signature you cannot attach to a face is a signature you will end up double-checking.

PLAN CARD 127W

Friday, three days after, and the first thing anybody says on walking into the base is that it makes no sense. Upper Town, on the granite bench, has parapets in the street and almost every building standing. The Flats, 1100 metres away on the same side of the same fault, have sand in the gutters, a six-storey block leaning at 8 degrees and a wharf that moved 2 metres. Marisol Okonkwo will not sign an assessment programme until she can say why. Elena Navarro thinks she already knows and wants a borehole to prove it. Today you read the two records side by side, work out what the ground did to the shaking on its way up, and decide which half of town gets the assessors first.

BRIEFING 18W

Upper Town lost its chimneys. The Flats lost its streets. Both are the same distance from the source.

WHAT YOU DECIDE 16W

Work out why the same shaking wrecked one half of Kestrel Bay and spared the other.

1.1 The same event, recorded twice

SEISMIC NETWORK · A ROOM · CHOICE

SCENE 44W

Inês Cardoso has the vault record from the bench and the field station record from Bay Road. Same event, same minute, 1100 metres apart. One trace is short and sharp, the other is three times taller and goes on for half a minute longer.

GUIDE 64W

All four options explain two records that disagree. They differ in what they blame: the instrument, the ground, the distance, or the timing. Two of them are checkable from the scene. Same event, same minute, 1100 metres apart. Ask what 1100 metres can and cannot account for. Which one is right decides whether the Flats get a stronger design value or a service call.

ASK 9W

What does the difference between the two records mean?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

an instrument records the shaking where it stands, not the earthquake itself

VERDICT 80W

Soft ground can amplify and lengthen shaking in a particular period band. Here the 2 records give $A_{\text{soft}}/A_{\text{rock}} \approx 3$ for the strongest motion. That is a site effect, not a larger earthquake under the Flats. Buildings also have preferred periods. A rough rule is $T \approx 0.1N$ seconds for an N -storey building. If strong site motion overlaps a building period, resonance can increase response. The ratio is frequency-dependent, so one factor should not be applied to every motion measure.

TAKEAWAY 17W

A record describes one place, so two records of one earthquake can disagree and both be right.

1.2 Ground that people put there

GEOTECHNICAL · A PERSON · PROTOCOL

SCENE 34W

Navarro has an 1892 survey and a photograph from 1948. Where Bay Road now runs there was a tidal creek, and the port was pumped into place out of the dredger over four summers.

GUIDE 57W

Four descriptions of ground, and four behaviours under shaking. Pair them by asking two things of each: how loose is it, and can the water get out. Age is not the variable it looks like. One of these is the wettest and loosest of all. It is a line on an 1892 survey, invisible from the street.

ASK 10W

Say what the ground under the Flats is made of

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

ground can be natural or placed by people

VERDICT 79W

Hydraulic fill is deposited loose and commonly remains saturated. During shaking, grains try to rearrange while water cannot escape quickly. Effective stress is $\sigma' = \sigma - u$: as pore-water pressure u rises, the grain-to-grain stress σ' falls. If that contact stress becomes very small, the soil can lose much of its shear strength and behave like a fluid. The old creek line matters because it marks especially wet, loose fill, which is also where the sand ejecta appeared.

TAKEAWAY 18W

Liquefaction needs susceptible soil, enough saturation and strong cyclic loading; young hydraulic fill often supplies the first two.

1.3 Where the first day of assessment goes

STRUCTURAL ASSESSMENT · A ROOM · ALLOCATE

SCENE 38W

Six assessors are free for one day. 400 displaced households are waiting on decisions in the Flats. Marina Court and the Parade are already cordoned, while Upper Town is mostly occupied. The team board shows six assessor-days available.

GUIDE 71W

Six assessor-days is the whole pool, and each package you drag in spends some of it. Watch the answers list rather than the buildings: it shows which questions the current plan can answer, and every package you buy is a question you can answer and one you have given up. The cordons already in place cost nothing. Commit the plan when the answers you can give are the ones today needs.

ASK 21W

Can your allocation preserve a same-day re-occupancy decision for the waiting Flats households while accepting that some lower-value questions will wait?

BEHIND THE BUTTON 148W

Why the pool is the constraint. Six assessors for one day is a fixed quantity of attention, and the Flats alone could absorb all of it. The decision is not what is worth inspecting — nearly everything is — but which inspections change what happens before tonight.

What makes an answer worth buying today. Four hundred households are waiting on a re-occupancy decision. An inspection that can produce that decision by this evening is worth more than one that produces a better answer next week, because the second one has already been made for them by the delay.

Why the existing cordons are on the board at zero. They are already in force, so they cost nothing to keep and they answer nothing new. They are there to be noticed: a plan that spends assessor-days re-establishing what is already closed has bought a question it had already answered.

TAKES AS READ

an assessment programme is a queue, and the order is a decision

VERDICT 81W

Assessment capacity is a finite pool, so every useful inspection has an opportunity cost. The Flats package can resolve an occupancy decision for hundreds of displaced households. Marina Court and the Parade already have protective controls in place, so extra detail there changes less today. Upper Town follow-up is useful but not essential to the immediate re-occupancy decision. The allocation should therefore preserve the Flats question first, then spend remaining capacity only where another answer is worth the assessor-days it consumes.

TAKEAWAY 18W

A finite inspection team should be spent on the questions whose answers can still change what happens today.

END OF THE DAY 18W

One earthquake produces as many shakings as there are grounds to shake, and the ground is the difference.

The number everybody has already heard

BEFORE THE DAY — TRIAL

Walk Upper Town before you placard anything

Three days after, and you have not been here before. Okonkwo, the structural engineer, takes every assessor round the near ground once on foot: the office, the structural bench, the materials lab. Nothing is placarded from a map, and the plan card names streets that mean nothing until you have stood in them and seen which way the ground falls.

PLAN CARD 113W

Saturday, and the town has two numbers for the same earthquake. 1 agency said 6.6 within 20 minutes; another said 6.8 6 hours later. Rei Tanaka can explain the difference in one sentence to an engineer and has spent the morning failing to explain it to a radio station. Funmi Adeyemi has to put something in a notice by four. Meanwhile Inês Cardoso has found that two field stations have been running on a clock that drifted since spring, which changes where the epicentre plots. Today you locate the earthquake properly, say what its magnitude is and what it is not, and write down what the difference between 6.6 and 6.8 actually means.

BRIEFING 14W

Two agencies published two magnitudes, and the difference is being read as a disagreement.

WHAT YOU DECIDE 14W

Pin the magnitude and the location down, and separate them from what people felt.

2.1 How far away it started

SEISMIC NETWORK · A ROOM · BALLPARK

SCENE 27W

The vault trace shows the P arrival at 04:12:19 and the S arrival at 04:12:32. Cardoso wants the distance before anybody draws a circle on a map.

GUIDE 68W

Five numbers, and two of them belong to the earthquake rather than to this station: the magnitude and the depth. Two others are clock readings, and only the gap between them carries distance. Ask of each whether it describes the source or the arrival. And note what one station can give: a radius, not a position. The circle still needs other stations before anybody draws on a map.

ASK 9W

Estimate the distance from this station to the source.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

P waves travel faster than S waves through the same rock

VERDICT 77W

Both waves leave together, but the P wave travels faster. Their arrival-time gap therefore grows with distance from the source. For shallow crust, about eight kilometres of source distance corresponds to each second of S-minus-P separation. The 13-second gap gives about 104 km. One station supplies only a radius, so the source still lies somewhere on a circle. A network combines several station distances to locate the source region rather than mistaking one distance for a position.

TAKEAWAY 20W

The gap between the two arrivals grows with distance, so one station gives a distance and three give a place.

2.2 Why the number moved

HAZARD & FORECASTING · INCIDENT BASE · A PERSON · BALLPARK

SCENE 31W

Tanaka has both agency bulletins on the desk. The early one used the first stations to report; the later one used the whole network and the long-period part of the record.

GUIDE 69W

The later solution has a fault model behind it: an area and an average slip. Build the seismic moment from the rigidity, the area and the slip, and the verdict turns that into a magnitude. Watch the units — the area is given in square kilometres and the moment needs square metres, so one of the tiles is the conversion already done and another is the same area unconverted.

ASK 15W

From the later solution's fault model, what seismic moment does the 6.8 estimate correspond to?

BEHIND THE BUTTON 162W

What the moment is made of. Rigidity times ruptured area times average slip. It is a physical quantity with units of newton metres, and it is what a long-period record measures — which is why a bulletin written from the first few seconds of short-period data can be revised once the whole waveform has arrived.

Why the scale is logarithmic. Magnitude is two thirds of the base-ten logarithm of the moment, less a constant, so one step of 0.2 in magnitude is a factor of about two in moment. Nothing about the earthquake changed between the bulletins; the estimate of how much fault moved by how much did.

Why this matters for the notice. A town told the number went from 6.6 to 6.8 hears that the earthquake grew. What actually happened is that the second estimate saw twice as much moment as the first one could, and the difference is a property of networks and waveforms rather than of the ground.

TAKES AS READ

a magnitude is computed from records, so more records can change it • magnitude as a logarithmic scale — taken as read

VERDICT 134W

Seismic moment is $M_0 = \mu AD$ — rigidity times ruptured area times average slip. The later solution's model is 600 km² of fault, which is 6.0×10^8 m², slipping an average of 1.1 m through crust with a rigidity of 3.0×10^{10} Pa, so $M_0 = 3.0 \times 10^{10} \times 6.0 \times 10^8 \times 1.1 \approx 2.0 \times 10^{19}$ N·m. Moment magnitude is $M_w = \frac{2}{3} \log_{10} M_0 - 6.06$, and $\frac{2}{3} \times 19.30 - 6.06 = 6.80$. The first bulletin's 6.6 is $10^{(1.5 \times 0.2)} \approx 2$ times smaller in moment, which is exactly what a rapid short-period estimate misses: the long-period part of the record is where a large slow rupture shows up. So the revision is a statement about network coverage and waveform bandwidth, and the fault moved once.

TAKEAWAY 21W

A magnitude revision means the estimate changed as the available waveform and network coverage improved; the earthquake itself did not change.

2.3 A notice somebody acts on

PUBLIC SAFETY · A ROOM · SEQUENCE

SCENE 30W

Adeyemi has 200 words and an audience that has heard 2 numbers and a rumour about a bigger 1 coming. She wants to know what the notice has to contain.

GUIDE 70W

All four of these will be in the notice, so nothing is being left out. Ask instead which one somebody can act on while standing in the street, and which is a fact they will repeat later. Two hundred words are read in the order they are printed. A revision from 6.6 to 6.8 changes nothing about what to do outside this building, and a rate is not a prediction.

ASK 9W

Say it in a way that survives being repeated

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a public notice is read once, quickly, by somebody who is tired · p and S waves, and what the gap between them measures — taken as read

VERDICT 78W

People act on the first thing they can use, so the action goes first. The reason comes next because it helps the instruction survive retelling. The magnitude revision belongs after the immediate action and explanation. It matters scientifically, but it does not change what someone should do outside this building. The aftershock forecast comes last and must be framed as a rate. That keeps a useful probability statement from being repeated as a prediction of a particular event.

TAKEAWAY 18W

A notice earns its place by changing what somebody does, and everything else in it competes with that.

A magnitude is a measurement with a method attached, and revising it is the method working.

What the placard is claiming

BEFORE THE DAY — FOLLOW

Mbeki walks the liquefaction line once

The boundary between the granite bench and the fill is not on any map at the accuracy anybody needs, and Mbeki, the geotechnical field technician, can see it in the ground. She walks it once, this morning, stopping where the surface changes. Stay with her — far enough back to see what she is pointing at, close enough not to lose which way she turned at the ejecta.

PLAN CARD 108W

Sunday, and the first placards are being argued about. Alan Whitcombe's team got through 400 buildings in 2 days, which is fast, and Whitcombe is the first to say what that speed bought and what it cost. A green placard on a house in the Flats means an engineer stood outside it for six minutes. Yvette Delacroix has residents asking whether green means safe, and Peter Ives at the hospital wants to know why his yellow has not moved in three days. Today you work out what a rapid assessment actually inspects, what a placard colour is claiming, and which buildings have to be looked at again properly.

BRIEFING 18W

400 buildings were placarded in 2 days by 6 people. Somebody has to say what that is worth.

WHAT YOU DECIDE 12W

Learn what a rapid assessment can honestly say, and what it cannot.

3.1 Six minutes outside a building

STRUCTURAL ASSESSMENT · A ROOM · CASEBOOK

SCENE 33W

Whitcombe's form has 11 boxes and a place for the colour. He filled 400 of them in 2 days, and he has kept a second list of everything he did not look at.

GUIDE 71W

The left column is four things about a building. The right column says whether six minutes outside it could have reached them. Ask of each: could somebody see this from the footpath? A form with a colour on it will be read later as a claim about the whole building. So what was not looked at has to be recorded as carefully as what was, and Whitcombe kept that second list.

ASK 7W

Say what a rapid assessment actually samples

BEHIND THE BUTTON 181W

Why explanations rather than labels. Naming a finding is not accounting for it, and a plausible-sounding mechanism attached to the wrong observation is the commonest way a wrong story survives. Committing an explanation to one clue means claiming it accounts for that clue specifically and not for its neighbour, which is where the two come apart.

How to use the one-each rule. The explanations are a set to be distributed, not a list to be sampled, so every join constrains the rest. Settling the two you are confident of can decide the remaining pair by elimination. Where it does not, two explanations are still competing for one clue, and that competition is the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every explanation used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

an inspection covers what the inspector could see

VERDICT 82W

A rapid assessment is designed to screen many buildings quickly after a disaster. It can identify gross leaning, collapse hazards, obvious structural distress and accessible life-safety problems. It cannot establish the condition of concealed connections, inaccessible rooms, foundations or every member in the load path. The discipline is therefore two-part: record what was observed and record what could not be observed. A green placard is useful only when later users do not silently expand its scope beyond the inspection that produced it.

TAKEAWAY 17W

A rapid assessment is triage: it records observed hazards and access limits, not a complete structural evaluation.

3.2 Green does not mean safe

PUBLIC SAFETY · A PERSON · CHOICE

SCENE 32W

Delacroix has three residents outside with green placards and one question between them: is my house all right? Adeyemi will not say yes and needs a sentence that is not a dodge.

GUIDE 60W

All four options are things a resident might hear as reassurance. They differ in how much they claim. Ask of each whether the rapid inspection could have established it. Three residents are outside asking whether their house is all right. A sentence that promises more than the inspection did is not kindness, and the placard will be quoted back later.

ASK 10W

What does a green placard entitle somebody to be told?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

people read a colour as a verdict

VERDICT 83W

The public message should be specific. The rapid inspection found no observed condition requiring an occupancy restriction in the areas that could be assessed. The form also records what the team could not inspect. That is different from promising the building is undamaged or fully evaluated. It also does not guarantee safety in every aftershock. The limitation is scientifically useful because it tells the owner what the inspection actually established and which unresolved areas may still need follow-up before a particular use resumes.

TAKEAWAY 20W

A placard is an occupancy decision based on a defined inspection, not a warranty that every hidden part is undamaged.

3.3 The ones worth a second visit

HAZARD & FORECASTING · INCIDENT BASE · A ROOM · TRIAGE

SCENE 36W

The field roster locks in ten minutes. Two engineers are free tomorrow, four addresses are on Tanaka's board, and each has somebody waiting for an answer. Once the roster closes, the other three wait another day.

GUIDE 62W

All four buildings deserve an answer and only two engineers are free. Ask of each two things: what did the first visit fail to see, and what is the cost of being wrong. An empty red building is already restricted, so a second visit changes little today. An anxious owner needs a conversation, which is not the same resource as an engineer-day.

ASK 7W

Which building gets the detailed evaluation first?

BEHIND THE BUTTON 194W

Why first is a different question from most important. When several calls compete for the same hour, what a choice is worth is not its own importance but the difference between doing it now and doing it later. A serious problem that will be no worse in an hour costs nothing to defer. A smaller one that closes a door costs everything behind that door.

What to look for in the options. Two things separate them: which is still changing, and which is a precondition for the others. A situation that is deteriorating has a cost per hour attached to it, and a step that unblocks the rest multiplies the value of the hours after it. Everything else is a preference about where to start.

Why only one answer is marked. In the situation these options describe all of them eventually happen; what is being tested is the head of the queue, because that is where the reasoning is visible. The verdict names what each of the others was waiting on, which is worth reading even when the choice was right — the ordering behind the first place is the rest of the answer.

TAKES AS READ

a second inspection is a scarce thing to spend

VERDICT 83W

A second inspection is valuable when the first one left an important condition unseen and the consequence of being wrong is high. The hospital is worst on both counts. 90 patients are inside, and the unresolved plant room is exactly the item preventing a clearer placard. The red warehouse is empty and already restricted, so another visit changes little today. The two green buildings have lower consequence. Their owners still deserve answers, but that can begin with communication rather than a scarce engineer-day.

TAKEAWAY 18W

Re-inspect where the first look was least able to see and the consequence of being wrong is largest.

3.4 The same event, recorded twice

SEISMIC NETWORK · A ROOM · CALLBACK · CHOICE

SCENE 44W

Inês Cardoso has the vault record from the bench and the field station record from Bay Road. Same event, same minute, 1100 metres apart. One trace is short and sharp, the other is three times taller and goes on for half a minute longer.

GUIDE 64W

All four options explain two records that disagree. They differ in what they blame: the instrument, the ground, the distance, or the timing. Two of them are checkable from the scene. Same event, same minute, 1100 metres apart. Ask what 1100 metres can and cannot account for. Which one is right decides whether the Flats get a stronger design value or a service call.

ASK 9W

What does the difference between the two records mean?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

an instrument records the shaking where it stands, not the earthquake itself

VERDICT 80W

Soft ground can amplify and lengthen shaking in a particular period band. Here the 2 records give $A_{\text{soft}}/A_{\text{rock}} \approx 3$ for the strongest motion. That is a site effect, not a larger earthquake under the Flats. Buildings also have preferred periods. A rough rule is $T \approx 0.1N$ seconds for an N -storey building. If strong site motion overlaps a building period, resonance can increase response. The ratio is frequency-dependent, so one factor should not be applied to every motion measure.

TAKEAWAY 17W

A record describes one place, so two records of one earthquake can disagree and both be right.

END OF THE DAY 20W

A placard states what an inspection covered, and the ones that go wrong are the ones read as covering more.

The building that looks worst

PLAN CARD 120W

Monday, and Marina Court is on the front page. Six storeys of flats sitting over at eight degrees above the marina, photographed from the water, and the photograph has done more to frighten this town than anything the office has said. Elena Navarro has been under it: the raft foundation is uncracked, the frame is plumb within itself, and the columns show nothing. Bram Halvorsen wants it demolished this week because of what it is doing to people. Marisol Okonkwo will not sign a demolition of a sound building on the strength of a photograph, and today she is right. Today you work out what actually failed, what the lean costs, and what a demolition would and would not fix.

BRIEFING 19W

Marina Court is leaning eight degrees and its structure is undamaged. The queue outside it is not about engineering.

WHAT YOU DECIDE 12W

Separate what frightens people from what is actually holding the building up.

4.1 The raft is fine and the building is not

GEOTECHNICAL • A ROOM • CHOICE

SCENE 34W

Navarro's survey: the raft is intact, level within itself, and rotated. 1 side has gone down 340 millimetres and the other has come up 90. The sand fans are worst on the low side.

GUIDE 50W

Four options, and the survey above has to be explained whole. Ask of each option how many of its four readings it covers. The intact raft. The level within itself. The rotation. And where the sand came up. A frame that failed would distort rather than rotate as one piece.

ASK 5W

Which explanation fits every observation?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a foundation spreads a building's weight into the ground • stress, strain and the yield point — taken as read

VERDICT 86W

The observations point to support beneath the raft rather than distortion within the frame. The raft is intact and the building rotated almost as a rigid body; one edge settled while the other heaved, with fresh sand ejecta concentrated on the low side. During liquefaction, excess pore pressure can sharply reduce effective stress and bearing resistance, allowing settlement and lateral movement without the frame itself yielding. That does not prove every detail, but it is the hypothesis that explains the survey, ejecta and undamaged frame together.

TAKEAWAY 10W

A structure can fail without anything in the structure breaking.

4.2 What eight degrees does

STRUCTURAL ASSESSMENT · A PERSON · CHOICE

SCENE 38W

Okonkwo has the column loads. At eight degrees the weight above each column no longer lands over its centre, and the offset grows with height up the building. The survey sheet and column plan are open beside him.

GUIDE 65W

All four options are true things about a leaning building. They differ in whether they name a load path or a feeling. Ask of each whether an engineer could put a number on it from the survey sheet and the column plan. A column carries weight best along its own axis. Eight degrees moves that weight off the axis, and the offset grows with height.

ASK 10W

What is the engineering objection to leaving Marina Court standing?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a leaning column carries its load off-centre

VERDICT 81W

A column is efficient when its gravity load stays close to its axis. Once the building leans, the weight above acts with an offset, creating additional bending moment; as lateral displacement grows, the gravity load can amplify that demand through P-Δ effects. The engineering question is therefore not whether eight degrees looks frightening, but whether the measured geometry, member capacities and continued ground stability leave adequate margin. The undamaged frame is encouraging evidence, not permission to ignore the new load path.

TAKEAWAY 19W

The lean matters because gravity now acts with eccentricity, creating $P-\Delta$ effects and extra bending demand in the frame.

4.3 What knocking it down would fix

PUBLIC SAFETY · A ROOM · VALUE

SCENE 36W

Halvorsen wants Marina Court demolished this week. Delacroix has 12 neighbouring buildings on the same reclaimed ground and 5 investigation-days before the cordon review. Four proposed uses of that window are written across the site map.

GUIDE 61W

Five investigation-days, four proposals, and a demolition somebody wants this week. Open each proposal to see what decision its result could change. Marina Court is already cordoned, so evidence about that one building changes nothing that is not already decided — the twelve neighbours on the same reclaimed ground are where the open decision is. Commit the window you would spend.

ASK 12W

Which investigation buys information that generalises beyond the one building already cordoned?

BEHIND THE BUTTON 148W

What is already decided and what is not. Marina Court is cordoned; nobody is going back in this week whatever a test says. The open question is whether the ground condition that damaged it requires action on the twelve buildings around it, and that is a question about the ground rather than about the building.

Why the most visible damage is the trap. A proposal to instrument the worst-damaged structure produces vivid results and answers a question that has been closed. It is the most defensible-sounding way to spend five days and learn nothing that changes a decision.

What generalising means here. A campaign of cone tests and boreholes across the Flats measures the fill itself, which every one of the thirteen buildings sits on. One result then bears on twelve open decisions instead of one closed one — the same money, an order of magnitude more use.

TAKES AS READ

a decision can treat a symptom or a cause

VERDICT 81W

Demolition can remove Marina Court as a visible hazard, but it does not reveal whether the same ground condition extends beneath nearby buildings. A corridor investigation can. CPTs, boreholes and groundwater checks test the ground shared by several sites while the existing cordon controls the known building hazard. Extra lean monitoring or a one-building underpinning concept answer narrower questions. Value of information therefore depends on the decision the evidence can change, not on how dramatic or precise the proposed work sounds.

TAKEAWAY 26W

The best next measurement is the one that can change the decision you still have to make, not the one that studies the most visible damage.

4.4 How far away it started

SEISMIC NETWORK · A ROOM · CALLBACK · BALLPARK

SCENE 27W

The vault trace shows the P arrival at 04:12:19 and the S arrival at 04:12:32. Cardoso wants the distance before anybody draws a circle on a map.

GUIDE 68W

Five numbers, and two of them belong to the earthquake rather than to this station: the magnitude and the depth. Two others are clock readings, and only the gap between them carries distance. Ask of each whether it describes the source or the arrival. And note what one station can give: a radius, not a position. The circle still needs other stations before anybody draws on a map.

ASK 9W

Estimate the distance from this station to the source.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

P waves travel faster than S waves through the same rock

VERDICT 77W

Both waves leave together, but the P wave travels faster. Their arrival-time gap therefore grows with distance from the source. For shallow crust, about eight kilometres of source distance corresponds to each second of S-minus-P separation. The 13-second gap gives about 104 km. One station supplies only a radius, so the source still lies somewhere on a circle. A network combines several station distances to locate the source region rather than mistaking one distance for a position.

TAKEAWAY 20W

The gap between the two arrivals grows with distance, so one station gives a distance and three give a place.

END OF THE DAY 18W

A building can fail without breaking, and the thing that failed may be the ground it stands on.

The one everybody assumed was fine

PLAN CARD 109W

Tuesday, and the school is the quiet problem. Bay Road School came through with a green placard, its classrooms look untouched, and the district would like 400 children back in it next week. Alan Whitcombe placarded it on day one from outside and wrote on his second list that the gymnasium was locked. The gym is a single-storey hall with a long-span roof, no internal walls, and heavy precast panels for its side walls. Marisol Okonkwo has the keys this morning. Today you work out what a long-span hall does in an earthquake, what holds its walls to its roof, and whether 400 children go into it on Monday.

BRIEFING 17W

Bay Road School is green, term starts in a week, and nobody has been inside the gym.

WHAT YOU DECIDE 14W

Look at a green building properly, and find what a walk-past could never see.

5.1 A hall with nothing in the middle

STRUCTURAL ASSESSMENT · A ROOM · CHAIN

SCENE 36W

Bay Road School's gym has a long roof, heavy precast wall panels and no interior bracing line. The structural drawings are pinned across a table. One panel connection tested poorly, but the larger members look undamaged.

GUIDE 61W

Build the path the sideways force actually takes, in order, from the moving mass to the ground. Then name the one required transfer that would break the path if it went. The largest member is not automatically the answer: a path is only as good as its weakest handover, and the readings on each link tell you which handover that is.

ASK 12W

Which link governs the panel load path when the building shakes sideways?

BEHIND THE BUTTON 176W

What a lateral system is. In an earthquake the mass of the building wants to keep going sideways while the ground moves under it. That inertia has to be carried by something, handed on, and eventually delivered to the foundation. Each of those handovers is a connection, and a connection that cannot carry its share ends the path there.

Why a gym is the hard case. A long roof with heavy precast panels and no interior bracing line has very few paths available: the roof has to act as a horizontal plate and hand its load to the end walls, and the panels have to be tied into that plate. Remove one tie and the panel is a free-standing wall carrying its own inertia.

Why undamaged members are not reassurance. The beams and columns can be in perfect condition while the connection between them is not, and a connection is exactly what the drawings say least about. One panel connection tested poorly, which is a statement about the class of connection rather than about that panel.

TAKES AS READ

a horizontal force has to travel through a building to reach the ground

VERDICT 85W

Earthquake inertia does not disappear because the visible structural members are large. A base-shear relation such as $V = C_s W$ estimates a total lateral demand, but that force still needs a continuous path to the ground. For the gym panel, inertia crosses the panel-to-roof tie, enters the roof diaphragm, reaches the resisting end wall or frame, and then reaches the foundation. The small tie is therefore load-bearing in the literal sense: if it opens, the rest of the strong path cannot collect that panel's force.

TAKEAWAY 19W

A lateral system works only when every required transfer carries force continuously from the moving mass to the foundation.

5.2 What the tested anchors actually carried

MATERIALS & TESTING · A PERSON · BALLPARK

SCENE 31W

Kirsten Sørensen has pulled two cast-in anchors from a spare panel in the yard. The design drawing says 40 kilonewtons each. The first came out at 24, the second at 27.

GUIDE 67W

Five numbers, and two of them belong to other questions: the roof span and the number of panels. One is what the drawing promised rather than what the wall got. Ask of each whether it describes what the anchors carried or what the wall asks of them. And note what two pulls cannot carry: a ratio is not a factor of safety, whatever it comes out at.

ASK 10W

Estimate the observed capacity-to-demand ratio for the two tested anchors.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a designed capacity is a claim to be checked

VERDICT 103W

Two anchors were pulled. They averaged 25.5 kN against a demand of 31 kN, so what was measured is about 0.82 of what the wall asks of them. The anchors are short. But 0.82 is not a factor of safety, and calling it one claims far more than two pulls can carry: a real check also has to hold the scatter between anchors, the way the joint lets go when it lets go, and the margins the code puts on both. None of that has to be settled to act. Restrict the gym, test enough anchors to mean something, and design the repair.

TAKEAWAY 26W

Two pull tests can show a serious capacity shortfall; they do not by themselves establish a code-level factor of safety for every anchor in the building.

5.3 400 children on Monday

PUBLIC SAFETY · A ROOM · CHOICE

SCENE 38W

400 children are due back Monday. The classrooms are sound and the gym is not. Halvorsen has parents calling, the district has no spare campus, and the morning bus schedule depends on whatever answer comes off this table.

GUIDE 64W

All four options are defensible readings of one site with two different buildings on it. Ask of each whether it treats the site as one thing or as parts. The classrooms and the gym have different evidence behind them. Four hundred children arrive Monday and there is no spare campus. A decision that covers both buildings equally is ignoring half of what was measured.

ASK 9W

What should happen to Bay Road School on Monday?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a building can be used in part

VERDICT 84W

The useful decision is component-specific. The classroom block has no identified structural deficiency, while the gym has heavy panels attached by connections whose tested capacity is below the estimated demand. If the gym can be physically isolated from classroom access and egress, the classrooms can reopen while the hall remains closed and the connections are repaired. If routes or falling hazards overlap, that separation has to be solved first. Partial occupancy is not optimism; it is a controlled use tied to the actual defect.

TAKEAWAY 17W

The useful question is which parts can be occupied, not whether the site is open or shut.

5.4 Ground that people put there

GEOTECHNICAL · A ROOM · CALLBACK · PROTOCOL

SCENE 34W

Navarro has an 1892 survey and a photograph from 1948. Where Bay Road now runs there was a tidal creek, and the port was pumped into place out of the dredger over four summers.

GUIDE 57W

Four descriptions of ground, and four behaviours under shaking. Pair them by asking two things of each: how loose is it, and can the water get out. Age is not the variable it looks like. One of these is the wettest and loosest of all. It is a line on an 1892 survey, invisible from the street.

ASK 10W

Say what the ground under the Flats is made of

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

ground can be natural or placed by people

VERDICT 79W

Hydraulic fill is deposited loose and commonly remains saturated. During shaking, grains try to rearrange while water cannot escape quickly. Effective stress is $\sigma' = \sigma - u$: as pore-water pressure u rises, the grain-to-grain stress σ' falls. If that contact stress becomes very small, the soil can lose much of its shear strength and behave like a fluid. The old creek line matters because it marks especially wet, loose fill, which is also where the sand ejecta appeared.

TAKEAWAY 18W

Liquefaction needs susceptible soil, enough saturation and strong cyclic loading; young hydraulic fill often supplies the first two.

END OF THE DAY 18W

The dangerous building is often the one nobody looked at, because nothing about it invited a second look.

The hospital, and what waiting costs

BEFORE THE DAY — HUNT

Six props went out and the log has four

Every temporary prop on the Parade has to be accounted for before the cordon can be reduced, because a prop that nobody can find is either holding something up or lying in a street where a lorry will hit it. Four are where the log says. Find the rest.

PLAN CARD 118W

Wednesday, and the hospital has been yellow for four days. Peter Ives has 90 patients on the ground floor and the 2 upper floors closed, the district's surgery is being done 60 kilometres away, and 2 people have already been transferred who would not have been. Marisol Okonkwo wants the plant room opened and the shear walls cored before she signs anything. Bram Halvorsen says every day of that is a day of care not delivered, and this morning he is right: the evidence Okonkwo wants will take four days and will not change what the building is. Today you work out what is actually unresolved, what evidence would change the decision, and what the waiting is costing.

BRIEFING 17W

90 patients, 3 days of yellow, and an argument about whether more evidence is worth more time.

WHAT YOU DECIDE 13W

Resolve a yellow placard on the one building the district cannot do without.

6.1 The part nobody has seen

STRUCTURAL ASSESSMENT · A ROOM · ATTEST

SCENE 40W

90 patients remain on the ground floor while 2 upper floors sit empty. The hospital is still yellow after three days. Frame, cladding and stair are documented. The roof plant room remains locked, with two full water tanks above it.

GUIDE 65W

Each claim on the list is signed or checked by somebody. Open the backing before you spend anything: some claims rest on an inspection of the thing they describe, and some rest on a document, an assumption or a locked door. You have one physical verification. Hold the claims the evidence does not support, and spend the visit where being wrong would cost the most.

ASK 11W

Which critical claim still needs physical verification on the current building?

BEHIND THE BUTTON 162W

What a record is and is not. A signature says somebody took responsibility for a statement. It does not say the condition was observed, and on a building three days after an earthquake those two come apart constantly: drawings are checked, cladding is looked at from the street, and a locked plant room is written up from the last inspection before the shake.

Why the tanks matter. Two full water tanks are mass, high up, in a room nobody has entered since the earthquake. If their supports have moved, the consequence lands on the floor below — where ninety patients are. The claim about that room is therefore the one whose being wrong costs the most, whatever its paperwork says.

Why one check rather than four. Verification costs time the hospital does not have while it stays yellow. Choosing which claim to check is the actual professional judgement, and it is a judgement about consequence rather than about which record looks flimsiest.

TAKES AS READ

a building's parts can be cleared separately

VERDICT 82W

A yellow placard should reduce to specific unresolved conditions, not a general feeling that more inspection is always safer. The frame, cladding and main stair already have direct observations behind their claims. The roof plant room does not. Its full water tanks add substantial mass high in the building, and nobody has seen their restraints. With one verification visit, repeating a comfortable backed inspection wastes the chance to resolve the one critical claim that still lacks physical evidence on the present configuration.

TAKEAWAY 22W

A signed or checked record is useful only to the extent that it is backed by observation of the condition it claims.

6.2 What four more days buys

HAZARD & FORECASTING · INCIDENT BASE · A ROOM · CHOICE

SCENE 37W

Tanaka has the aftershock rate for the coming week. Ives has the transfer log. Halvorsen wants both of them stated as consequences rather than as principles. The review board meets again at the end of the week.

GUIDE 61W

All four options are ways of weighing four days. They differ in what they count. Ask of each whether it prices the delay, the evidence, or neither. The frame has already been inspected and the unresolved item is a roof plant room nobody has entered. So the real test is what this evidence could change, and coring answers a different question.

ASK 10W

How should the office weigh four more days of evidence-gathering?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a delay has consequences that can be counted

VERDICT 89W

Both sides of the decision have consequences and both can be estimated. Delay carries transfers, postponed care and operational strain. Additional testing is worthwhile when it has a realistic chance of changing the safety decision. Here the frame has already been inspected and the unresolved item is the unvisited roof plant room with heavy tanks; a four-day coring programme answers a different question. The disciplined test is therefore value of information: what decision could this evidence change, how likely is that, and what is paid while waiting for it?

TAKEAWAY 22W

A decision to wait is a decision with an outcome, and it has to be compared with the outcome of deciding now.

6.3 The instrument in the basement

SEISMIC NETWORK · A ROOM · BALLPARK

SCENE 34W

There is an accelerograph in the hospital basement, installed in 1998 and never used in anger. Cardoso has the record. It shows a peak of 0.31 g, and the building was designed for 0.35.

GUIDE 66W

Turn the recorded acceleration into the sideways force the frame actually had to carry, so the two numbers can be compared as forces rather than as fractions of gravity. The base shear is the seismic coefficient times the building's weight. One tile is the coefficient the building was designed to and another is the weight of one floor, which is not what the whole frame carries.

ASK 17W

What sideways force did the recorded motion put into the frame, against what it was designed for?

BEHIND THE BUTTON 155W

What a base shear is. The total horizontal force at the foundations, and the number a frame is proportioned against. It is the seismic coefficient — the design acceleration as a fraction of gravity, after the code's factors — multiplied by the weight the building brings to the shaking.

Why 0.31 against 0.35 is not a verdict. It is one input, measured directly, which is worth a great deal in a fortnight where most numbers are inferred. What a building feels also depends on which frequencies arrived, how long the shaking lasted, and how the frame and its parts are detailed. None of that is in a peak.

What the record cannot reach. It says nothing about the plant-room restraint, nothing about what the inspection found, and nothing about the parts of the building that respond at their own period rather than the ground's. A design value is not a threshold below which nothing breaks.

TAKES AS READ

a building can be instrumented, and the record is about that building · peak ground acceleration and what a building feels — taken as read · p and S waves, and what the gap between them measures — taken as read

VERDICT 202W

The hospital is four storeys, so its natural period is about $T \approx 0.1 N = 0.4 s$, and that is the period at which the design coefficient is read off the spectrum. Base shear is then $V = C_s \cdot W$: that coefficient times the weight the building brings to the shaking. The hospital's seismic weight is about 85 meganewtons, and the recorded peak of 0.31 g gives a coefficient of 0.31, so the frame saw roughly $0.31 \times 85 \approx 26$ MN. The design value of 0.35 g gives $0.35 \times 85 \approx 30$ MN. So the demand was about 12 per cent under what the frame was proportioned for, which is worth knowing and is not a clearance. Three things the number does not carry: which frequencies

arrived, how long the shaking went on, and how the building and its contents are detailed — the plant-room restraint responds to its own motion, not to the basement's. A design value is a proportioning input, not a line below which nothing breaks. What the record genuinely removes is the largest unknown of the fortnight: for once, the ground motion at a specific building was measured instead of inferred from a ratio.

TAKEAWAY 22W

A building-mounted accelerograph measures one part of the demand directly; it does not turn a code design value into a damage threshold.

6.4 The frame, or the finish

STRUCTURAL ASSESSMENT · A PERSON · BELT

SCENE 42W

Four hundred photographs came in from the rapid assessment and the panel sits at four. Some of what is in them is the structure. The rest is what was hung on it, and the two produce the same alarm in a caption.

GUIDE 47W

Two bins. Structural damage is to the things carrying load down to the ground — columns, beams, walls that hold something up, connections between them. Non-structural damage is to what the building carries: cladding, ceilings, partitions, services. Sort on whether the item is in the load path.

ASK 15W

Send each photograph to the bin that says whether it is in the load path.

BEHIND THE BUTTON 82W

Why the split decides the placard rather than the repair bill. A cracked partition is expensive and does not change whether the building stands up in the next aftershock. A cracked column does, and it can be behind a partition that looks fine.

Why non-structural damage still kills people. Falling cladding, a collapsed ceiling grid and a toppled bookcase all injure. They are dealt with by clearing and securing rather than by shoring, which is a different crew and a different day.

TAKES AS READ

some parts of a building hold it up and some are carried by it

VERDICT 149W

The placard question is whether the building will stand up to the next shake, and only what carries load answers it. Columns, beams, shear walls and their connections are the path down to the foundation, and damage to any of them changes capacity. A shattered partition, a collapsed ceiling grid, cladding hanging off a facade and a toppled bookcase are all carried *by* that path, and none of them makes the frame weaker. That is why the sort matters more than it sounds: the alarming photographs are mostly non-structural, and the photograph that decides the placard is often a hairline diagonal on a column behind an intact wall. Non-structural damage still injures people — falling cladding is the commonest injury in a moderate earthquake — so the answer is not that it does not matter. It is that it wants a different crew, a different urgency, and no shoring.

TAKEAWAY 14W

What decides occupancy is what is in the load path, not what looks worst.

6.5 A notice somebody acts on

PUBLIC SAFETY • A ROOM • CALLBACK • SEQUENCE

SCENE 30W

Adeyemi has 200 words and an audience that has heard 2 numbers and a rumour about a bigger 1 coming. She wants to know what the notice has to contain.

GUIDE 70W

All four of these will be in the notice, so nothing is being left out. Ask instead which one somebody can act on while standing in the street, and which is a fact they will repeat later. Two hundred words are read in the order they are printed. A revision from 6.6 to 6.8 changes nothing about what to do outside this building, and a rate is not a prediction.

ASK 9W

Say it in a way that survives being repeated

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a public notice is read once, quickly, by somebody who is tired • p and S waves, and what the gap between them measures — taken as read

VERDICT 78W

People act on the first thing they can use, so the action goes first. The reason comes next because it helps the instruction survive retelling. The magnitude revision belongs after the immediate action and explanation. It matters scientifically, but it does not change what someone should do outside this building. The aftershock forecast comes last and must be framed as a rate. That keeps a useful probability statement from being repeated as a prediction of a particular event.

TAKEAWAY 18W

A notice earns its place by changing what somebody does, and everything else in it competes with that.

Gathering more evidence is not free, and evidence that cannot change the decision costs without buying anything.

What the ground will do next

PLAN CARD 113W

Thursday, and the question in every room is when the aftershocks end. Rei Tanaka has the sequence: 210 events above magnitude 3 in 4 days, falling steadily. Bram Halvorsen wants a date he can put on a notice and Funmi Adeyemi has to write it. Marisol Okonkwo needs the rate for a different reason — a damaged building has a reduced capacity, and the chance of it being tested again is the only thing that makes a red placard urgent rather than tidy. Today you fit the decay, say what the forecast is and is not, and work out what a falling rate means for the buildings still standing with cracks in them.

BRIEFING 12W

41 aftershocks yesterday, 26 today. Everybody wants to know when it stops.

WHAT YOU DECIDE 11W

Turn an aftershock sequence into a rate somebody can plan around.

7.1 The rate, falling

HAZARD & FORECASTING · INCIDENT BASE · A ROOM · BALLPARK

SCENE 30W

Day 1 had 96 events above magnitude 3. Day 2 had 52, day 3 41, day 4 26. Tanaka's fit has the rate roughly halving as the elapsed time doubles.

GUIDE 63W

Five numbers, and two of them belong to other questions: the magnitude and the running total. One is day one, which is four doublings back rather than one. Ask of each whether it is a count, a time, or a ratio of times. And note what comes out: an expected rate, not a quota. Real days scatter either side of a fitted curve.

ASK 10W

Estimate the number of magnitude 3+ aftershocks on day 8.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a rate can fall smoothly without ever reaching zero

VERDICT 77W

A simple Omori-style model writes the aftershock rate as $n(t) = K/(c+t)^p$. When p is near one and c is small after the first day, doubling elapsed time roughly halves the central rate. Day 4 had 26 magnitude-3-plus events, so day 8 gives about $26 \times 4/8 = 13$. Real sequences scatter around that curve, and K , c and p are fitted with uncertainty. 13 is therefore an expected rate, not a quota or a scheduled event.

TAKEAWAY 18W

A simple Omori-style decay gives a central rate estimate; the real forecast also needs uncertainty around that curve.

7.2 A rate is not a schedule

SEISMIC NETWORK · A PERSON · BALLPARK

SCENE 36W

The notice draft says the aftershocks will have stopped by the end of the month. Cardoso points at the sentence and asks where the month came from. The sentence is due for the afternoon public bulletin.

GUIDE 64W

Put the fitted numbers through the decay law and see what day thirty actually gives. The rate falls as a power of the time since the mainshock, so pick the productivity, the time offset and the elapsed days, and read the answer as events per day. One tile is the exponent, which goes in the power and not in the denominator on its own.

ASK 17W

At day 30 after the mainshock, what does the fitted decay law give as the expected rate?

BEHIND THE BUTTON 180W

What the law says. The rate of aftershocks per day falls roughly as one over the elapsed time raised to a power near one, with a small offset that keeps the first hours finite. It is a decay, not a countdown: the curve approaches zero and never arrives, which is why no end date can be read off it.

Why a fitted rate is not a schedule. The number that comes out is an expectation over an interval — so many per day on average. Actual counts scatter around it, some days three and some days none, and a rate of a tenth of an event per day does not promise the tenth day is the one.

Why size is a separate question. The decay law describes how often, and the magnitude distribution describes how large. A falling rate lowers the chance of a large aftershock in the same proportion as a small one, and rules nothing out. Publishing a month is therefore two mistakes: an end where there is a fade, and a date where there is a rate.

TAKES AS READ

a forecast states what is expected on average · aftershock decay as a power law — taken as read

VERDICT 156W

The Omori–Utsu law is $n(t) = K / (c + t)^p$, and the fitted values for this sequence are $K = 84$, $c = 0.2$ days and $p = 1.05$. At $t = 30$ days the denominator is $30.2^{1.05} \approx 35.6$, so the expected rate is $84 \div 35.6 \approx 2.4$ to 2.7 events a day above magnitude 2, depending on how the fit is rounded. That is an expectation over an interval, and daily counts will land either side of it. What the curve cannot do is reach zero: a power law fades, and the day it 'stops' does not exist. Publishing a month therefore makes two errors at once — an end date where the mathematics gives a fade, and a certainty where it gives a rate. The right sentence names a number per week and a

range, and says nothing about the largest event, because how often and how large are different distributions.

TAKEAWAY 22W

A rate says how often, not when or how large, and writing it as a date makes a promise nobody can keep.

7.3 Damaged, and waiting to be asked again

STRUCTURAL ASSESSMENT · A ROOM · CHOICE

SCENE 37W

Okonkwo has 11 buildings with cracked shear walls that are still standing. Each has less capacity than it had a week ago, and the sequence has not finished. The shoring list is pinned beside the new forecast.

GUIDE 60W

All four options give a reason the eleven buildings stay urgent. They differ in what is changing: the hazard, the buildings, or both. Ask of each whether the falling aftershock rate contradicts it. Two of these say the shaking will get worse, and the forecast says otherwise. What has changed since Friday is on the other side of the comparison.

ASK 12W

Why does a falling aftershock rate still leave these 11 buildings urgent?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a damaged element carries less than it used to

VERDICT 83W

The hazard is declining, but the buildings are not the same buildings they were before the mainshock. Cracked shear walls may have lost capacity, so a smaller aftershock can now produce additional distress. That does not mean every magnitude-5 event will damage them or that the exact remaining capacity is known. It means the acceptable demand has changed. The 11 buildings therefore stay urgent because the sequence still supplies repeated loading while their structural reserve is reduced and has not yet been restored.

TAKEAWAY 18W

Damage lowers the capacity, so the same aftershock that was survivable last week may not be this week.

7.4 Green does not mean safe

PUBLIC SAFETY · A ROOM · CALLBACK · CHOICE

SCENE 32W

Delacroix has three residents outside with green placards and one question between them: is my house all right? Adeyemi will not say yes and needs a sentence that is not a dodge.

GUIDE 60W

All four options are things a resident might hear as reassurance. They differ in how much they claim. Ask of each whether the rapid inspection could have established it. Three residents are outside asking whether their house is all right. A sentence that promises more than the inspection did is not kindness, and the placard will be quoted back later.

ASK 10W

What does a green placard entitle somebody to be told?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

people read a colour as a verdict

VERDICT 83W

The public message should be specific. The rapid inspection found no observed condition requiring an occupancy restriction in the areas that could be assessed. The form also records what the team could not inspect. That is different from promising the building is undamaged or fully evaluated. It also does not guarantee safety in every aftershock. The limitation is scientifically useful because it tells the owner what the inspection actually established and which unresolved areas may still need follow-up before a particular use resumes.

TAKEAWAY 20W

A placard is an occupancy decision based on a defined inspection, not a warranty that every hidden part is undamaged.

END OF THE DAY 20W

An aftershock forecast is a rate with a range, and the useful question is what it does to a decision.

The material, and what it can still carry

PLAN CARD 117W

Friday, and the argument at the base has moved from buildings to numbers. The public library is a 1970s concrete frame with diagonal cracking in 3 ground-floor columns, and 3 engineers have looked at it and used the words moderate, serious and superficial.

Duarte Ferreira points out that none of those is a number, and that his laboratory can produce one in a day. Kirsten Sørensen has already cut four cores and refused a fifth for arriving without a label saying which column it came from. Today you take a sample properly, turn it into a strength, compare it with what the design assumed, and decide what the building may be used for while that is happening.

BRIEFING 13W

The library's columns look cracked. Nobody has said what they can still hold.

WHAT YOU DECIDE 14W

Get a number for the strength of a damaged building instead of an opinion.

8.1 A number, with a method attached

MATERIALS & TESTING · A ROOM · BALLPARK

SCENE 37W

4 cores, 100 mm across, cut from the cracked columns. Sørensen caps and crushes them: 28, 31, 24 and 30 megapascals. The 1974 drawing calls for 25. The four labelled specimens are still on the testing bench.

GUIDE 58W

Work the third core's strength out from the machine's own numbers rather than reading it off the report. Stress is the crushing force divided by the area the force acted on, and the area is the circle the core presents — not its diameter. One tile is that diameter, left for somebody who divides by the wrong thing.

ASK 17W

The third core failed at 244 kN. What compressive strength is that, and what does it settle?

BEHIND THE BUTTON 170W

Why the area and not the diameter. A stress is a force spread over a surface, so the number underneath has to be an area in square metres. A 100 mm core presents a circle of about 0.00785 m^2 , and using 0.1 instead gives an answer twelve times too small with no unit to warn you.

What the strain is doing. The machine also records shortening: 0.42 mm over a 200 mm gauge length is a strain of 0.0021, which is dimensionless because it is a length over a length. Stress against strain up to the yield point is what the material's stiffness is, and it is the pair that makes a stress–strain curve.

What a core does not settle. A member's capacity depends on the force it actually carried, on its reinforcement and on how well that steel confines the concrete — none of which travels in a cylinder. Four cores near the specified strength argue against grossly weak concrete and say nothing about a cracked column's remaining capacity.

TAKES AS READ

concrete strength is measured by crushing a sample of known size

VERDICT 150W

Stress is $\sigma = F / A$. The core is 100 mm across, so the area carrying the load is $\pi \times 0.05^2 \approx 0.00785 \text{ m}^2$, and 244 kN over that area is $244,000 \div 0.00785 \approx 31 \text{ MPa}$. The machine's other channel gives the strain: $\epsilon = \Delta L / L$, and 0.42 mm of shortening over a 200 mm gauge is 0.0021. Together those two numbers are a point on the stress–strain curve, and the ratio between them up to yield is the stiffness. What none of it reaches is the column: capacity depends on the load path, the reinforcement and the confinement that steel provides, and a cylinder carries none of them. Four cores centred on the specified 25 MPa rule out grossly weak concrete as the explanation for the cracking. That is one input removed from the list, which is worth having and is not the answer.

TAKEAWAY 16W

Core tests characterize sampled concrete; they do not certify the residual capacity of the cracked columns.

8.2 Which way the cracks run

STRUCTURAL ASSESSMENT · A PERSON · CHOICE

SCENE 37W

The cracks run diagonally, in both directions, crossing each other in an X across the middle third of each column. They are fine, numerous and evenly spaced.

Photographs of all three columns are clipped to the sheet.

GUIDE 60W

Four options, and the pattern has to be explained in full. The cracks are diagonal, in both directions, crossing in the middle third. They are fine, numerous and evenly spaced. Ask of each option what pattern it would leave instead. Crushing and settlement do not reverse direction. And note the limit: a pattern names a mechanism, not a remaining capacity.

ASK 5W

Which explanation fits the pattern?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a crack runs across the direction of the tension that opened it · stress, strain and the yield point — taken as read

VERDICT 80W

Diagonal cracking is consistent with principal tension produced by shear, and crossed diagonals fit load reversals during earthquake shaking. The pattern is therefore much more consistent with cyclic shear demand than with simple long-term settlement or axial crushing. Fine, distributed cracks can indicate that reinforcement is controlling crack widths rather than allowing one brittle split, but that is not the same as proving the column is safe. Crack pattern identifies a likely mechanism; residual capacity still needs an engineering check.

TAKEAWAY 19W

The shape and repetition of damage can narrow the failure mechanism, but pattern recognition alone cannot establish remaining strength.

8.3 Part of a building, for part of a purpose

PUBLIC SAFETY · A ROOM · SEQUENCE

SCENE 38W

Monday's public meeting has 200 residents registered and no backup venue. The library is the only hall large enough. Three ground-floor columns are cracked, and the route plan has to be settled before the doors can be advertised.

GUIDE 66W

All four of these will happen, so the order is what is being decided. Ask of each what has to be settled before it can mean anything. A number of people is only enforceable once the permitted area is defined. A review date on a restriction is what stops it drifting into permanence. Two hundred residents are registered for Monday and there is no other hall.

ASK 9W

Decide what the building may be used for meanwhile

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

occupancy can be limited by number, by area or by time

VERDICT 83W

Partial occupancy is an engineering boundary, not a hopeful label. The meeting can use the hall only if occupied routes and seating stay outside the hazard zone.

Damaged columns must also be isolated or supported as required. The temporary use must be defined in writing. A restriction saying only 'part of the building' is too vague to enforce. The plan should name the permitted area, excluded area, route and duration. That preserves usable space without silently widening the claim made by the assessment.

TAKEAWAY 21W

Restricted use is a real option between open and shut, and it has to name what the restriction is protecting against.

8.4 Why the number moved

HAZARD & FORECASTING · INCIDENT BASE · A ROOM · CALLBACK · BALLPARK

SCENE 31W

Tanaka has both agency bulletins on the desk. The early one used the first stations to report; the later one used the whole network and the long-period part of the record.

GUIDE 69W

The later solution has a fault model behind it: an area and an average slip. Build the seismic moment from the rigidity, the area and the slip, and the verdict turns that into a magnitude. Watch the units — the area is given in square kilometres and the moment needs square metres, so one of the tiles is the conversion already done and another is the same area unconverted.

ASK 15W

From the later solution's fault model, what seismic moment does the 6.8 estimate correspond to?

BEHIND THE BUTTON 162W

What the moment is made of. Rigidity times ruptured area times average slip. It is a physical quantity with units of newton metres, and it is what a long-period record measures — which is why a bulletin written from the first few seconds of short-period data can be revised once the whole waveform has arrived.

Why the scale is logarithmic. Magnitude is two thirds of the base-ten logarithm of the moment, less a constant, so one step of 0.2 in magnitude is a factor of about two in moment. Nothing about the earthquake changed between the bulletins; the estimate of how much fault moved by how much did.

Why this matters for the notice. A town told the number went from 6.6 to 6.8 hears that the earthquake grew. What actually happened is that the second estimate saw twice as much moment as the first one could, and the difference is a property of networks and waveforms rather than of the ground.

TAKES AS READ

a magnitude is computed from records, so more records can change it • magnitude as a logarithmic scale — taken as read

VERDICT 134W

Seismic moment is $M_0 = \mu AD$ — rigidity times ruptured area times average slip. The later solution's model is 600 km² of fault, which is 6.0×10^8 m², slipping an average of 1.1 m through crust with a rigidity of 3.0×10^{10} Pa, so $M_0 = 3.0 \times 10^{10} \times 6.0 \times 10^8 \times 1.1 \approx 2.0 \times 10^{19}$ N·m. Moment magnitude is $M_w = \frac{2}{3} \log_{10} M_0 - 6.06$, and $\frac{2}{3} \times 19.30 - 6.06 = 6.80$. The first bulletin's 6.6 is $10^{(1.5 \times 0.2)} \approx 2$ times smaller in moment, which is exactly what a rapid short-period estimate misses: the long-period part of the record is where a large slow rupture shows up. So the revision is a statement about network coverage and waveform bandwidth, and the fault moved once.

TAKEAWAY 21W

A magnitude revision means the estimate changed as the available waveform and network coverage improved; the earthquake itself did not change.

END OF THE DAY 21W

A word like serious is an opinion; a strength with a method behind it is something the next engineer can use.

The cordon nobody can lift

BEFORE THE DAY — CANVASS

Who actually spent the night in their own bed

The occupancy figures the office is working from came from Saturday, and they decide which buildings get assessed first. Ask along the affected streets until enough answers have come back to say whether people went to the hall as instructed or came home once the shaking stopped.

PLAN CARD 115W

Saturday, and the cordon around the Flats is a week old. It was drawn on day one for falling masonry, kept on day three for the gas main, and is being kept today because nobody has said it can come down. Yvette Delacroix has a list of 400 households, 60 of whom are sleeping in the sports hall, and a question the office has not answered: what has to be true for people to go home? Bram Halvorsen is losing patience and Marisol Okonkwo will not be hurried. Today you find out what the cordon is actually protecting against street by street, what each part of it costs, and where the line should be tonight.

BRIEFING 16W

11 streets have been closed for 6 days, and the reason is now 3 different reasons.

WHAT YOU DECIDE 13W

Weigh what a cordon protects against what it costs the people inside it.

9.1 Three reasons wearing one line

PUBLIC SAFETY · A ROOM · ATTEST

SCENE 41W

Shopkeepers are waiting at the tape with deliveries and keys in hand. Delacroix walks the cordon with the boundary log. Three stretches have written reasons. The southern leg is still closed, but nobody can point to the hazard it is controlling.

GUIDE 71W

Each stretch of cordon is a claim that a hazard is there now. Open the evidence behind each one before spending anything: some rest on an inspection, some on a hazard that has since been removed, and one has no written reason at all. You can recheck one stretch physically. Hold the claims the evidence does not support, and choose the recheck where opening the wrong street would cost the most.

ASK 16W

Which current exclusion claim most needs to be checked against the physical condition on the ground?

BEHIND THE BUTTON 165W

A cordon is a claim, not a decision. Tape is put up for a reason, and the reason is a statement about a present condition — falling glass, an unstable parapet, ground that may still move. Conditions change and reasons expire, so a boundary nobody has revisited is a claim nobody has retested.

What the cost of a stale cordon is. Shopkeepers are at the tape with keys and deliveries. Every day a stretch stays closed without a hazard behind it, the cordon as a whole loses authority, and the stretches that do control real hazards get walked around. Over-closing is not the safe error it looks like.

Why the unwritten leg is not automatically the answer. A stretch with no logged reason may still be controlling a genuine hazard that somebody saw on day one and never wrote down. The point of the check is to find out, which is why the panel makes you spend the verification rather than infer from the paperwork.

TAKES AS READ

a boundary can be drawn for more than one reason

VERDICT 80W

A cordon can contain several hazards, each with its own condition for removal. The Parade segment has an overhead-masonry basis. The gas-main segment depends on utility status. Ferry Street is an access restriction for heavy emergency vehicles. The southern leg is different. The map still excludes people, but the log contains no recorded hazard supporting that claim. With one field check before the map is reissued, the unbacked segment deserves verification before a restriction with direct support is checked again.

TAKEAWAY 22W

Every cordon segment is a claim about a current hazard, and each claim should remain traceable to evidence that can be rechecked.

9.2 Soft ground, three days on

GEOTECHNICAL · A PERSON · CHOICE

SCENE 35W

Thandi Mbeki has been on Ferry Street with a hand auger. The sand fans have dried, the water table has dropped back, and the surface bears a person easily and a truck not at all.

GUIDE 62W

Four options describing the same street, from ruined to recovered. Ask of each which observation supports it and which contradicts it. The fans have dried and the water table has dropped, so the excess pressure has gone. The surface takes a person and not a truck. A dry surface says nothing about heavy loads, and drained is not the same as unchanged.

ASK 9W

What is the defensible status of Ferry Street now?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

liquefied ground regains strength as pore pressure dissipates

VERDICT 79W

Liquefaction needs high pore pressure during shaking. Afterward, that excess pressure drains away and grain contacts recover. The ground can still be badly changed. Settlement, lateral spreading, ejecta, local voids and uneven stiffness may remain. Some loose sand may even become denser, so calling the whole deposit 'looser' is too simple. A dry surface also says little about truck support. Ferry Street is disturbed ground with heavy-load capacity unverified. Susceptible saturated layers could liquefy again in a strong aftershock.

TAKEAWAY 21W

Liquefaction is transient, but the ground left behind can have settlement, voids and uncertain bearing capacity even after pore pressure dissipates.

9.3 A condition, not a date

HAZARD & FORECASTING · INCIDENT BASE · A ROOM · CHOICE

SCENE 39W

Halvorsen wants a date. Delacroix wants a condition. Tanaka points out that a date makes a promise about the ground and a condition makes a promise about the office. The next resident update is due before the office closes.

GUIDE 0W

ASK 6W

What should the office commit to?

TAKES AS READ

a commitment can be tied to a date or to an event · peak ground acceleration and what a building feels — taken as read · p and S waves, and what the gap between them measures — taken as read

VERDICT 87W

A date promises something about the ground that the office cannot control. A condition promises what evidence the office will require before changing the restriction. Publishing those conditions turns an open-ended wait into visible progress. Residents can see what remains: utility clearance, access checks, targeted assessment or repair. The aftershock rate is a poor release trigger because it does not measure the local hazard on a specific street. The commitment should therefore name the observations and actions that will justify reopening, not a day selected for reassurance.

TAKEAWAY 18W

Tie a decision to a condition somebody can check, and it stays honest whatever the ground does next.

9.4 Where the first day of assessment goes

STRUCTURAL ASSESSMENT · A ROOM · CALLBACK · ALLOCATE

SCENE 38W

Six assessors are free for one day. 400 displaced households are waiting on decisions in the Flats. Marina Court and the Parade are already cordoned, while Upper Town is mostly occupied. The team board shows six assessor-days available.

GUIDE 71W

Six assessor-days is the whole pool, and each package you drag in spends some of it. Watch the answers list rather than the buildings: it shows which questions the current plan can answer, and every package you buy is a question you can answer and one you have given up. The cordons already in place cost nothing. Commit the plan when the answers you can give are the ones today needs.

ASK 21W

Can your allocation preserve a same-day re-occupancy decision for the waiting Flats households while accepting that some lower-value questions will wait?

BEHIND THE BUTTON 148W

Why the pool is the constraint. Six assessors for one day is a fixed quantity of attention, and the Flats alone could absorb all of it. The decision is not what is worth inspecting — nearly everything is — but which inspections change what happens before tonight.

What makes an answer worth buying today. Four hundred households are waiting on a re-occupancy decision. An inspection that can produce that decision by this evening is worth more than one that produces a better answer next week, because the second one has already been made for them by the delay.

Why the existing cordons are on the board at zero. They are already in force, so they cost nothing to keep and they answer nothing new. They are there to be noticed: a plan that spends assessor-days re-establishing what is already closed has bought a question it had already answered.

TAKES AS READ

an assessment programme is a queue, and the order is a decision

VERDICT 81W

Assessment capacity is a finite pool, so every useful inspection has an opportunity cost. The Flats package can resolve an occupancy decision for hundreds of displaced households. Marina Court and the Parade already have protective controls in place, so extra detail there changes less today. Upper Town follow-up is useful but not essential to the immediate re-occupancy decision. The allocation should therefore preserve the Flats question first, then spend remaining capacity only where another answer is worth the assessor-days it consumes.

TAKEAWAY 18W

A finite inspection team should be spent on the questions whose answers can still change what happens today.

END OF THE DAY 22W

A cordon is a decision that is remade every day, and one kept out of habit is a decision nobody is making.

The photograph that changed the meeting

PLAN CARD 115W

Sunday, and the office is on the back foot. A resident photographed a cracked column in the basement car park of a green-placarded apartment block on Ferry Street and gave it to the paper. The photograph is real, the crack is real, and the block has 90 flats with people back inside it. Alan Whitcombe placarded it green from outside on day one and never saw the basement. Funmi Adeyemi has a public meeting at six. Marisol Okonkwo has two hours and no access key. Today you work out what the photograph does and does not establish, get into the basement, and decide what to say at six o'clock about a process that missed it.

BRIEFING 17W

A resident's photograph of a cracked column is on the front page, and the building is green.

WHAT YOU DECIDE 11W

Deal with evidence that arrives from outside the process, in public.

10.1 A real crack, and an unknown building

STRUCTURAL ASSESSMENT · A ROOM · CHOICE

SCENE 37W

The photograph has already been passed around town. It shows a basement column with a horizontal crack, pipework and no daylight. There is no scale, no building number and no way to tell which column it is.

GUIDE 66W

Four options, and they differ in how much a photograph can carry. There is no scale, no building number and no earlier picture. Ask of each option what it would need that the image does not have: a width, an age, a location, a cause. Accepting the crack as real is not the same as knowing what it means, and it is already all over town.

ASK 5W

What has the photograph established?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a photograph records what was in front of the lens

VERDICT 81W

Once the image is accepted as genuine, the crack is evidence, but its scope is narrow. The photograph has no scale, exact location, earlier comparison or useful view of the surrounding structure. It cannot establish the crack's width, age, importance or cause. It does establish that a basement area outside the rapid inspection contains a real feature worth checking. The response is therefore to locate it and gather context quickly. The photograph alone neither justifies evacuation nor supports dismissing the observation.

TAKEAWAY 15W

An authenticated photograph can establish an observation while leaving scale, location, age and mechanism unresolved.

10.2 Down there with a torch

MATERIALS & TESTING · A PERSON · CHOICE

SCENE 37W

In the basement: the crack is 0.4 mm and horizontal, a metre above the floor. Five more columns along the row carry one that looks the same. The original construction drawings are open on a crate nearby.

GUIDE 61W

Four explanations, and one fact constrains all of them. Six columns in a row carry the same feature at the same height and the same width. Ask of each option whether it would produce that repetition. Six independent failures rarely agree to a millimetre. The drawings are open on a crate, which is where a shared construction detail would show up.

ASK 8W

What is the best-supported explanation to test first?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a defect that repeats in a pattern usually has a construction cause

VERDICT 85W

A horizontal feature of nearly identical height and width across six columns strongly suggests a construction detail shared by all six, such as the top of a concrete lift.

That is more likely than six independent shear failures producing the same geometry. But earthquake damage can also exploit construction joints, so 'construction joint' is a hypothesis, not an automatic clearance. Check drawings or pour records, inspect displacement and spalling across the joint, and compare with undamaged columns before deciding whether the earthquake changed its condition.

TAKEAWAY 22W

When the same feature repeats at the same place in many members, look first for a shared detail before assuming separate failures.

10.3 The meeting

PUBLIC SAFETY · A ROOM · SEQUENCE

SCENE 40W

200 people are already in the hall. One photograph has been forwarded all over town. The office has to explain what it found tonight, what it did not see on day one, and what happens next before anyone goes home.

GUIDE 66W

All four of these have to be said tonight, so nothing is being withheld. Ask which one the hall came to hear, and which one it will only accept afterwards. Two hundred people are already seated and the photograph has been forwarded all over town. An explanation offered before the finding reads as a defence, and the admission is the part that has to be plain.

ASK 5W

Say it at six o'clock

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

trust is affected by how a process handles being wrong

VERDICT 88W

The finding goes first because that is what everyone came to hear. Its meaning follows: the repeated horizontal crack fits a construction joint better than new earthquake shear damage. Then the office should state the process gap plainly. The basement was not inspected, the omission was recorded, and nobody acted on the list. The final step is the change in practice. This order keeps the meeting focused on evidence first. It also turns acknowledgment of a process failure into a specific corrective action rather than a defensive argument.

TAKEAWAY 22W

Say what was found, what it means and what the process missed, in that order, and commit to the part you control.

10.4 The water table, while the trench is open

GEOTECHNICAL · A ROOM · HOLD

SCENE 40W

The trench for the temporary shoring has to stand open for an afternoon in ground that liquefied nine days ago. The wellpoints hold the water down. If the water comes back up the sides do not wait to be asked.

GUIDE 54W

Hold the water table below the trench floor, inside the band on the gauge. The band narrows as the afternoon goes on, because each hour the sides stand open they lose a little of what was holding them. The wellpoint pumps are your control, and every inflow keeps arriving until the pumps answer it.

ASK 13W

Hold the water table below the trench floor while the shoring goes in.

BEHIND THE BUTTON 141W

Why the water is what matters. The sand is held together by the pressure between grains, and that is the weight above less the water pressure between them. Let the water rise and the grains stop pressing on each other while nothing about the sand has changed.

Why an inflow is a rate. Rain on the fill, a leaking main, the tide in the harbour — each of them keeps feeding water in for as long as it lasts, so the table does not settle at a new level. It keeps rising until pumping matches the inflow.

Why the margin shrinks. An open trench face creeps: it loses a little strength every hour it stands, so the same rise in water is a bigger fraction of what is left by the end of the afternoon than it was at the start.

TAKES AS READ

water pressure between grains reduces the strength of sandy ground

VERDICT 143W

Sandy ground holds together because the grains press on one another, and that pressure is the weight above less the water pressure between them. Raise the water and the grains stop pressing, though nothing about the sand has changed — which is the same mechanism that liquefied this ground nine days ago, arriving slowly instead of in forty seconds. Every disturbance here is a rate. Rain on the fill keeps arriving, the leaking main keeps leaking, the tide keeps coming in, so the table does not settle at a new level and a pump started briefly buys nothing. Set the pumps to match the inflow and leave them there. The band narrows because an open face creeps: it loses a little strength every hour it stands, so the rise that was tolerable at one o'clock is most of the remaining margin by four.

TAKEAWAY 17W

A trench is held open by a pressure balance, and the balance is held by a pump.

10.5 The ones worth a second visit

HAZARD & FORECASTING · INCIDENT BASE · A ROOM · CALLBACK · TRIAGE

SCENE 36W

The field roster locks in ten minutes. Two engineers are free tomorrow, four addresses are on Tanaka's board, and each has somebody waiting for an answer. Once the roster closes, the other three wait another day.

GUIDE 62W

All four buildings deserve an answer and only two engineers are free. Ask of each two things: what did the first visit fail to see, and what is the cost of being wrong. An empty red building is already restricted, so a second visit changes little today. An anxious owner needs a conversation, which is not the same resource as an engineer-day.

ASK 7W

Which building gets the detailed evaluation first?

BEHIND THE BUTTON 194W

Why first is a different question from most important. When several calls compete for the same hour, what a choice is worth is not its own importance but the difference between doing it now and doing it later. A serious problem that will be no worse in an hour costs nothing to defer. A smaller one that closes a door costs everything behind that door.

What to look for in the options. Two things separate them: which is still changing, and which is a precondition for the others. A situation that is deteriorating has a cost per hour attached to it, and a step that unblocks the rest multiplies the value of the hours after it. Everything else is a preference about where to start.

Why only one answer is marked. In the situation these options describe all of them eventually happen; what is being tested is the head of the queue, because that is where the reasoning is visible. The verdict names what each of the others was waiting on, which is worth reading even when the choice was right — the ordering behind the first place is the rest of the answer.

TAKES AS READ

a second inspection is a scarce thing to spend

VERDICT 83W

A second inspection is valuable when the first one left an important condition unseen and the consequence of being wrong is high. The hospital is worst on both counts. 90 patients are inside, and the unresolved plant room is exactly the item preventing a clearer placard. The red warehouse is empty and already restricted, so another visit changes little today. The two green buildings have lower consequence. Their owners still deserve answers, but that can begin with communication rather than a scarce engineer-day.

TAKEAWAY 18W

Re-inspect where the first look was least able to see and the consequence of being wrong is largest.

END OF THE DAY 22W

Evidence does not stop being evidence because of who found it, and a process that cannot absorb it will be worked around.

A quiet day on the bench

PLAN CARD 115W

Monday, and for the first time in ten days nothing is on fire. The Flats have their programme, the hospital is open, the school is running in its classrooms, and the office has a day with no crisis in it. Marisol Okonkwo spends those days on the half of town nobody is asking about, because Upper Town was cleared quickly on the strength of standing up, and standing up is not the same as undamaged. Alan Whitcombe walks the Parade with her. Today you look at masonry properly, work out what holds a heavy old wall to the floors behind it, and decide what is worth doing to buildings that are not currently anybody's emergency.

BRIEFING 14W

Upper Town has been assessed, cleared and forgotten. Somebody should look at it properly.

WHAT YOU DECIDE 12W

Spend a day on the half of town nobody is arguing about.

11.1 Walls that fall outward

STRUCTURAL ASSESSMENT · A ROOM · CHAIN

SCENE 40W

Owners are asking when the Parade can reopen. Every parapet fell outward into the street, while the buildings behind them remain standing. Timber floors are still connected. The fallen masonry is tagged by frontage, and the barriers are still up.

GUIDE 53W

Follow the sideways force from the shaking ground up through the building and name the link that gave way. Each step carries a reading saying what restraint it can actually provide. The heaviest element on the board is not automatically the one that failed, and neither is the first step in the path.

ASK 13W

Trace the sideways force through the frontage and name the link that failed.

BEHIND THE BUTTON 163W

Why out of plane. Masonry is strong in compression and weak in tension, so a wall pushed sideways cannot hold itself up by bending. It stays standing only while something ties it back at intervals — a floor, a roof, or a strap into the diaphragm. Where the tie stops, the wall is a free cantilever.

Why the parapet and not the wall. The wall below is held by the timber floors, which are still connected and did their job. The parapet projects above the roof line with nothing above it and nothing behind it, so the whole horizontal force it collects has to be resisted at its base, in bending, by mortar.

Why this is the useful kind of failure. It is local, repeatable and explained by one missing connection, so the retrofit targets the tie rather than the building. A frontage whose parapet is anchored back to the roof structure is a different proposition from one whose whole wall needs work.

TAKES AS READ

masonry is strong in compression and weak in tension

VERDICT 148W

Horizontal shaking gives every part of the frontage an inertia force proportional to its mass, and that force has to travel somewhere. Below roof level it goes into the timber floors, which are connected and stiff enough in their own plane to carry it to the cross walls and down to the foundations — every one of those links had capacity to spare. Above roof level the path stops. A parapet has nothing tying its top back and nothing above it, so the entire force it collects must be resisted at its base by mortar in bending, which is the one thing unreinforced masonry cannot do. That is why every parapet on the Parade went and the walls below them stayed. It also decides the repair: the missing link is a connection, so the retrofit is an anchor into the roof structure rather than work on the wall.

TAKEAWAY 20W

Unreinforced masonry fails out of plane, so the free top of a wall goes first and the tied part stays.

11.2 What you can learn without breaking it

MATERIALS & TESTING · A PERSON · VALUE

SCENE 37W

Ferreira has four test proposals for the Parade's listed brickwork. The scaffold permit allows one small trial location this week, and owners will not accept large-scale destructive sampling. A tie-back design still needs a defensible capacity number.

GUIDE 62W

Four tests are on the table and the permit allows one trial location. Open each card: it says what the test measures and how much listed fabric it consumes. The tie-back design needs a defensible pull-out capacity, so the test worth buying is the one that measures that property — not the one that produces the most precise number about something else.

ASK 10W

Which measurement answers the capacity question the retrofit depends on?

BEHIND THE BUTTON 164W

What the design actually needs. A tie-back is only as good as what it is anchored into, so the number the drawing needs is how much force an anchor in this brickwork can carry before it pulls out. Compressive strength, mortar composition and moisture content are all real properties of the wall, and none of them is that number.

Why precision is not value. A laboratory can give three significant figures on a property the design does not use, and the report will look better than a crude pull-out test with wide scatter. The question is which measurement changes the decision, which is the same question VALUE asks in every game that has one.

Why the fabric cost is on the card. This is listed brickwork: every test damages something that cannot be replaced, and the owners have already refused large-scale sampling. A test's cost here is not money but heritage, which makes choosing well a conservation decision as much as an engineering one.

TAKES AS READ

a test can be destructive, semi-destructive or neither

VERDICT 80W

The design question is whether an anchor installed in this old wall can transfer the required force. A trial-anchor pull-out test measures that property directly in the existing masonry and at the intended embedment. Brick compressive strength, rebound hardness and mortar condition can all describe parts of the wall, but none is the same as anchor pull-out capacity. The listed façade also makes disturbance a real cost. The best test therefore couples decision relevance with the smallest adequate physical intervention.

TAKEAWAY 18W

A precise test is valuable only when it measures the property that the pending design decision actually needs.

11.3 Before the next one

SEISMIC NETWORK · A ROOM · SCIENCETANK

SCENE 32W

Nothing on the Parade is dangerous today. The same parapets, on the same walls, will be here for the next earthquake, and the recurrence interval on this fault is about 140 years.

GUIDE 83W

Every parapet on the Parade came down and no wall behind one did. Tying back costs about £8,000 a building while the scaffolding stands, and about £25,000 once it is gone. A detailed assessment of one building takes three weeks and produces no strengthening. The grant scheme opens in eleven months and is oversubscribed in its first week. Lightweight replacement needs the parapets down first and a consented design, which is next summer at the earliest. The last event was six days ago.

ASK 6W

Decide what is worth doing now

BEHIND THE BUTTON 193W

Why the whole spread is graded. Funding is not a vote for one idea. A portfolio says what you think is likely, what is worth hedging against, and what is not worth pursuing at all — and the last two are where most of the information is. Backing the right proposal while quietly funding a bad one is a worse answer than it looks. That is why the small numbers count as much as the big one.

What the three numbers are for. Thirty-five is what makes a lead a lead: below it you have hedged rather than chosen. Fifteen is the most that can sit on unsupported work before it stops being a rounding error. Past that it is a second opinion nobody argued for. Twenty is the floor under a line of work you have already called strong, because funding it too thin to finish spends the money and buys nothing.

Why there is a floor on the total. Points held back are not caution; they are a decision not to decide, taken with somebody else's money and somebody else's deadline. The floor is what forces the panel to say something.

TAKES AS READ

a hazard that has occurred will occur again

VERDICT 84W

Tying parapets back has an unusually strong case: the failure mode occurred repeatedly, the retrofit acts directly on it, and scaffolding already in place makes the marginal cost much lower. The quoted 140-year recurrence interval is not a promise of 140 quiet years after this event; earthquakes do not reset a calendar. That statistic belongs in long-term hazard context, not in the argument for waiting. Act now because the defect is known, the consequence is falling masonry, and the cheapest access window is open.

TAKEAWAY 25W

Fix a repeatedly observed, low-cost life-safety failure mode while access is already in place; do not let an average recurrence interval become a countdown clock.

11.4 The raft is fine and the building is not

GEOTECHNICAL • A ROOM • CALLBACK • CHOICE

SCENE 34W

Navarro's survey: the raft is intact, level within itself, and rotated. 1 side has gone down 340 millimetres and the other has come up 90. The sand fans are worst on the low side.

GUIDE 50W

Four options, and the survey above has to be explained whole. Ask of each option how many of its four readings it covers. The intact raft. The level within itself. The rotation. And where the sand came up. A frame that failed would distort rather than rotate as one piece.

ASK 5W

Which explanation fits every observation?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a foundation spreads a building's weight into the ground • stress, strain and the yield point — taken as read

VERDICT 86W

The observations point to support beneath the raft rather than distortion within the frame. The raft is intact and the building rotated almost as a rigid body; one edge settled while the other heaved, with fresh sand ejecta concentrated on the low side. During liquefaction, excess pore pressure can sharply reduce effective stress and bearing resistance, allowing settlement and lateral movement without the frame itself yielding. That does not prove every detail, but it is the hypothesis that explains the survey, ejecta and undamaged frame together.

TAKEAWAY 10W

A structure can fail without anything in the structure breaking.

END OF THE DAY 19W

The building that survived is the one you learn most from, because you can still see what saved it.

The record that was wrong all along

BEFORE THE DAY — EVADE

The developer who wants a green placard in the street

A developer has been in the office car park since eight wanting a verbal assurance about Marina Court, and Cardoso, the network technician, has said plainly that nothing is agreed anywhere but on the form. Keep clear of him and get your round done. A placard agreed in a car park is a placard nobody signed.

PLAN CARD 116W

Tuesday, and Inês Cardoso has found the thing nobody wanted found. The vault above Upper Town is the reference station — the rock record every comparison in this fortnight has been made against — and when she went up to service it she looked at the trench log from its installation in 1998. It is not founded on rock. It sits on 4 metres of weathered, broken granite that behaves like a stiff soil, and it has been amplifying by about 1.6 since the day it was built. Every amplification figure this office has quoted is therefore too small. Today you work out what that does to the fortnight's conclusions, and which of them still stand.

BRIEFING 22W

The vault on the bench is not on rock. It is on four metres of weathered granite, and that changes every comparison.

WHAT YOU DECIDE 11W

Re-read the fortnight after finding the instrument everyone trusted was mis-sited.

12.1 Measuring against the wrong zero

SEISMIC NETWORK · A ROOM · TRACE

SCENE 40W

Cardoso's five-day side-by-side check is finished, and the benchmark used since day one no longer stands unchanged. Navarro lays five fortnight conclusions beside the corrected station report. Some may survive untouched; others may force a redraw before the next briefing.

GUIDE 60W

Open a conclusion and the panel shows what it was computed from. Keep the ones that stand on a measurement of their own, and untick the ones whose chain runs through the corrected reference. Then name the source the failing ones share. Both halves are graded: throwing away an independent conclusion costs you as much as keeping a dependent one.

ASK 14W

Which conclusions actually inherit the bad reference, and which remain supported by independent evidence?

BEHIND THE BUTTON 169W

What a shared reference is. Every station's record is interpreted against an assumed site response, and one station's has been corrected after five days of side-by-side comparison. Anything computed through that station inherits the correction; anything measured elsewhere does not. The error is not in the instrument — it is in one number every dependent conclusion was divided by.

Why untangling matters more than caution. The safe-looking move is to withdraw the whole fortnight, and it is wrong twice over: it throws away conclusions that were never affected, and it tells the next briefing that nothing is known when several things are. A retraction is a claim too, and it should be as narrow as the evidence.

How to read a chain. A conclusion is supported if some path to it never touches the suspect source. That is a question about the graph rather than about the size of the numbers, which is why the panel makes you open each one instead of judging by how alarming it looks.

TAKES AS READ

a comparison is only as good as the thing compared against

VERDICT 103W

A ratio inherits the error of whatever sits under the line, and the panel gives you both terms: a published $\times 3.0$ and a reference station that reads $\times 1.6$ where $\times 1.0$ was assumed. The corrected value is $3.0 \times 1.6 \approx 4.8$ times competent rock, and the demand factor copied from that ratio carries the same dependency. The hospital basement accelerograph measured 0.31 g on its own instrument, the moment magnitude came off a multi-station solution and Marina Court was surveyed on the ground, so none of the three moves. Dependency tracing limits the correction to the conclusions that actually share the bad reference.

TAKEAWAY 22W

A shared reference error propagates only through conclusions that depend on it; independent measurements should not be discarded with the common source.

12.2 What still stands

GEOTECHNICAL · A PERSON · CASEBOOK

SCENE 33W

Navarro puts four of the fortnight's conclusions on the board and asks which of them leaned on the vault ratio and which did not. The corrected vault report is clipped beside the board.

GUIDE 63W

Four conclusions from the fortnight, and the reference station turns out to be founded on weathered granite. Trace each conclusion back to its evidence and ask whether the vault is in it. A survey, a heave and a field of ejecta do not need a reference station. Neither does a basement instrument. One bad denominator invalidates what divided by it and nothing else.

ASK 4W

Sort the fortnight's conclusions

BEHIND THE BUTTON 181W

Why explanations rather than labels. Naming a finding is not accounting for it, and a plausible-sounding mechanism attached to the wrong observation is the commonest way a wrong story survives. Committing an explanation to one clue means claiming it accounts for that clue specifically and not for its neighbour, which is where the two come apart.

How to use the one-each rule. The explanations are a set to be distributed, not a list to be sampled, so every join constrains the rest. Settling the two you are confident of can decide the remaining pair by elimination. Where it does not, two explanations are still competing for one clue, and that competition is the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every explanation used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a conclusion can rest on one measurement or on several

VERDICT 83W

Trace each conclusion to the evidence that supports it. The Flats amplification ratio directly used the vault as its reference, so it must be restated. Marina Court's diagnosis came from settlement, heave, ejecta and structural survey. The hospital's 0.31 g came from its own basement instrument. The parapet case rests on observed building failures and restraint details. Those conclusions may deserve other checks, but this reference-station error does not erase them. One bad denominator should not become an excuse to discard unrelated evidence.

TAKEAWAY 16W

A bad reference invalidates what was measured against it and leaves what was measured directly alone.

12.3 Five times, not three

HAZARD & FORECASTING · INCIDENT BASE · A ROOM · BALLPARK

SCENE 37W

If the Flats amplify by five rather than three, the shaking a future earthquake will deliver there is much larger than the figure in the district plan. The revised district map is waiting for a replacement number.

GUIDE 56W

Two instruments recorded the same earthquake, one on the Flats and one on rock above the town. The amplification is the ratio between what they measured, so pick the two peak accelerations and divide. One tile is the figure the district plan currently carries, which is the number being replaced rather than an input to it.

ASK 15W

From the two records of the same earthquake, what is the Flats' amplification over rock?

BEHIND THE BUTTON 150W

What the ratio is. Site amplification is the peak motion on soft ground over the peak motion on competent rock for the same event. It is dimensionless, so both readings have to be in the same units, and it is a property of the ground rather than of the earthquake.

Why one number is not a spectrum. Soft ground amplifies some frequencies far more than others, so a single ratio is a summary. It is the summary the district plan uses, which is why replacing three with five moves every design value that plan feeds — and why the full answer is a curve somebody still has to measure.

What the correction cannot touch. The placards came from what inspectors saw in buildings, and the cracks are already photographed. A revised ratio changes the shaking a future earthquake is expected to deliver, and rewrites nothing that has already been observed.

TAKES AS READ

design values are chosen from expected shaking · peak ground acceleration and what a building feels — taken as read

VERDICT 162W

Site amplification is $A_{\text{soft}} / A_{\text{rock}}$: the peak acceleration recorded on soft ground divided by the peak on competent rock for the same event. The Flats instrument recorded 0.41 g and the vault instrument on granite recorded 0.082 g, so the ratio is $0.41 \div 0.082 \approx 5.0$, against the 3 the district plan assumes. Both readings are in g, so the answer is dimensionless, which is what makes it a property of the ground rather than of this earthquake. Two things follow and one does not. The design values for rebuilding in the Flats rise, and the case for ground improvement gets much stronger. What does not change is anything already observed: the placards came from inspections of real buildings, and the magnitude of the earthquake was measured from waveforms, not inferred from a ratio. And one number is still only a

summary — amplification varies with frequency, so the Flats need a response curve before anything is designed against them.

TAKEAWAY 18W

Correcting site response changes forward-looking hazard and design assumptions; it does not rewrite damage that was observed directly.

12.4 What the teams are being sent to

PUBLIC SAFETY · A ROOM · SPOT

SCENE 38W

Buildings queue for assessment faster than the teams can reach them. The office works to one priority, and Okonkwo changes it as the fortnight does — after the aftershock, after the schools ask, after the cordon goes up.

GUIDE 53W

Send teams to what the current priority wants and leave the rest queued. The priority on the office board changes during the day and nobody announces it. What is scored is the buildings either side of a change, because they are the only ones that show whether the office is reading the board.

ASK 12W

Dispatch to the priority on the board, and keep watching the board.

BEHIND THE BUTTON 92W

Why the priority changes. Early it is occupied buildings, because people are in them. After an aftershock it is anything already placarded yellow, because their capacity has fallen and the same shake now does more. When the schools ask it is schools, because a decision is waiting on the answer rather than on the risk.

Why the changeover costs. A team sent under the old priority spends half a day on a building nobody is waiting for, and the building that the new priority named waits another night with people in it.

TAKES AS READ

an office decides which buildings are assessed first

VERDICT 116W

Three priorities run across the fortnight and each wants a different part of the queue. Occupied buildings, then anything already placarded yellow, then schools. A building can answer two at once, which is what makes the change cost real rather than notional, and the panel scores the window either side of each change because most of the queue is wanted by neither priority and is correctly left waiting by somebody who has read nothing. The office's own version is the hour after an aftershock: the priority has changed to the yellow placards, whose capacity has just fallen again, and the teams already in the vans are driving to the list that was current before the shake.

TAKEAWAY 15W

The cost of a withdrawn instruction is paid by whoever is still working to it.

12.5 Six minutes outside a building

STRUCTURAL ASSESSMENT · A ROOM · CALLBACK · CASEBOOK

SCENE 33W

Whitcombe's form has 11 boxes and a place for the colour. He filled 400 of them in 2 days, and he has kept a second list of everything he did not look at.

GUIDE 71W

The left column is four things about a building. The right column says whether six minutes outside it could have reached them. Ask of each: could somebody see this from the footpath? A form with a colour on it will be read later as a claim about the whole building. So what was not looked at has to be recorded as carefully as what was, and Whitcombe kept that second list.

ASK 7W

Say what a rapid assessment actually samples

BEHIND THE BUTTON 181W

Why explanations rather than labels. Naming a finding is not accounting for it, and a plausible-sounding mechanism attached to the wrong observation is the commonest way a wrong story survives. Committing an explanation to one clue means claiming it accounts for that clue specifically and not for its neighbour, which is where the two come apart.

How to use the one-each rule. The explanations are a set to be distributed, not a list to be sampled, so every join constrains the rest. Settling the two you are confident of can decide the remaining pair by elimination. Where it does not, two explanations are still competing for one clue, and that competition is the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every explanation used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

an inspection covers what the inspector could see

VERDICT 82W

A rapid assessment is designed to screen many buildings quickly after a disaster. It can identify gross leaning, collapse hazards, obvious structural distress and accessible life-safety problems. It cannot establish the condition of concealed connections, inaccessible rooms, foundations or every member in the load path. The discipline is therefore two-part: record what was observed and record what could not be observed. A green placard is useful only when later users do not silently expand its scope beyond the inspection that produced it.

TAKEAWAY 17W

A rapid assessment is triage: it records observed hazards and access limits, not a complete structural evaluation.

END OF THE DAY 22W

A reference that is not what it claims moves every measurement made against it, and finding that out is a good day.

Three things at once

PLAN CARD 114W

Wednesday, 15:40, and 3 things arrive together. A magnitude 5.1 aftershock has just shaken the town, which is the largest in a week. A water main has burst on Ferry Street and is pouring into ground everybody now knows is loose. And the district wants a decision on the school gymnasium by five o'clock, because contractors are holding a slot for Saturday. Marisol Okonkwo has four engineers, Bram Halvorsen is on the phone to the utility, and Funmi Adeyemi has three drafts of a notice nobody has approved. Today you decide what gets attention first, work out what the aftershock did to buildings already damaged, and say what a burst main does to fill.

BRIEFING 15W

An aftershock, a burst main and a deadline for the school, all inside an hour.

WHAT YOU DECIDE 9W

Rank three demands on one office in one afternoon.

13.1 Which of the three cannot wait

PUBLIC SAFETY · A ROOM · DELEGATE

SCENE 36W

Three calls land within minutes. A burst main is feeding water into the Flats. Several vulnerable buildings need checks after the M5.1 aftershock, and Bay Road School needs its reopening message issued. Four engineers are available.

GUIDE 66W

Three calls, four engineers, and one of the three you have to keep yourself. For each of the others, choose an owner who can take a real first action and the reading or event that brings the problem back to you. A handover with an owner and no return condition is not a handover. Then take the watch on the one that needs your own judgement.

ASK 27W

How do you divide the team so the worsening condition is stopped, the life-safety consequence is checked and the stable deadline does not consume the field response?

BEHIND THE BUTTON 167W

Which one to keep. One of these is getting worse while you decide, one has a life-safety consequence that has to be checked, and one is a deadline that is stable — the message will still need issuing in an hour. Keeping the stable one is the classic error: it is the easiest to finish, and finishing it costs the field response its hours.

Why a first action is required. "Watch it" is not a handover. An owner who cannot take a concrete step is a person who will come back for instructions at the worst moment, so the panel asks for the action they will take before they need to ask anybody.

Why a return condition is the other half. Delegation is not disposal: the problem comes back if it crosses a threshold, and naming that threshold now is what lets somebody else hold it without checking in. A burst main over liquefiable fill and a reopening message have very different triggers, and both need one.

TAKES AS READ

a problem still growing is different from one that has finished

VERDICT 76W

The burst main is actively worsening the ground condition, so utility isolation starts immediately. The M5.1 aftershock is over, but vulnerable or shored buildings may have new distress. That uncertainty needs engineering judgment before restrictions change. The school decision is already technically settled and mainly needs communication. Parallel work is therefore better than a single ranked list. Stop the growing input, keep direct attention on uncertain building consequences, and hand the stable message to the desk.

TAKEAWAY 28W

Good delegation couples every handoff to a first action and a condition for returning the problem, while scarce expert attention stays on the judgment that cannot be automated.

13.2 Wetting ground that has already failed

GEOTECHNICAL · A PERSON · BALLPARK

SCENE 33W

Mbeki is on Ferry Street with a standpipe. The water table under the road has risen three quarters of a metre since the main went, and the ground surface is soft again underfoot.

GUIDE 70W

Put a number on what the leak did. The grains are held together by the effective stress, which is the total weight above them less the water pressure between them. Pick the total stress at three metres and the pore pressure the risen water table now gives, and take one from the other. Two tiles are the pressures before the main burst, which is the comparison rather than the answer.

ASK 18W

At three metres down under Ferry Street, what is the effective stress now the water table has risen?

BEHIND THE BUTTON 148W

What effective stress is. The soil skeleton carries the difference between the total stress pressing down and the water pressure in the pores pushing the grains apart. Strength lives in that difference, which is why a saturated fill can be strong one day and behave like a liquid the next without a gram of it moving away.

What the leak did and did not do. A higher water table raises the pore pressure and takes the margin down. It liquefies nothing on its own — cyclic shaking is still required to build the pressure the rest of the way. What has changed is how much shaking that now takes.

Why Upper Town is different. On competent granite the same burst main is a plumbing problem. There is no loose saturated skeleton whose grain contacts can be unloaded by water pressure, so the mechanism has nothing to act on.

TAKES AS READ

liquefaction requires saturated ground

VERDICT 183W

Effective stress is $\sigma' = \sigma - u$. The fill weighs about 18 kN per cubic metre, so at three metres the total vertical stress is $18 \times 3 = 54$ kPa, and that has not changed. What has changed is the water. The table was 2.25 m down, giving a pore pressure at three metres of $9.81 \times 0.75 \approx 7.4$ kPa and an effective stress of $54 - 7.4 \approx 47$ kPa. It is now 1.5 m down, so $u = 9.81 \times 1.5 \approx 14.7$ kPa and $\sigma' = 54 - 14.7 \approx 39$ kPa. The skeleton is carrying about 16 per cent less than it was on Monday, and everything the ground can resist scales with that number. The leak has liquefied nothing by itself; cyclic shaking is still required to raise u the rest of the way to σ . It has simply moved the starting point closer, which is why the same aftershock forecast means something different on Ferry Street this week. On granite there is no loose saturated skeleton for the mechanism to act on at all.

TAKEAWAY 21W

Raising the water table reduces effective stress in susceptible fill, increasing liquefaction susceptibility during renewed shaking without causing liquefaction by itself.

13.3 A smaller shake against a weaker building

STRUCTURAL ASSESSMENT · A ROOM · CHOICE

SCENE 30W

The 5.1 was about a tenth of the mainshock's ground motion in the Flats. 11 buildings have been shored since Friday, and 2 of them have new cracking this afternoon.

GUIDE 54W

Four readings of the same afternoon. Ask of each what the shoring was for. It is temporary support, not a repair, so new cracking is not automatically a failure of it. What has changed since Friday is on the other side of the comparison. Eleven buildings have less capacity, and two of them responded.

ASK 11W

What do the two buildings with new cracking tell the office?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a shored building is being held rather than repaired · p and S waves, and what the gap between them measures — taken as read

VERDICT 84W

Shoring is temporary load support, not restoration of the damaged structural system. New cracking during a smaller aftershock therefore matters: it shows that the current combination of residual capacity, shoring and aftershock demand produced additional distress. That is a reason to maintain or tighten restrictions and perform a focused reassessment before deciding on strengthening, evacuation or a revised shore scheme. It does not by itself prove the shoring failed, identify the exact damaged mechanism, or tell you how close the building is to collapse.

TAKEAWAY 23W

New cracking during a smaller aftershock is evidence that residual capacity or detailing deserves escalation; it is not a complete diagnosis by itself.

13.4 What the tested anchors actually carried

MATERIALS & TESTING · A ROOM · CALLBACK · BALLPARK

SCENE 31W

Kirsten Sørensen has pulled two cast-in anchors from a spare panel in the yard. The design drawing says 40 kilonewtons each. The first came out at 24, the second at 27.

GUIDE 67W

Five numbers, and two of them belong to other questions: the roof span and the number of panels. One is what the drawing promised rather than what the wall got. Ask of each whether it describes what the anchors carried or what the wall asks of them. And note what two pulls cannot carry: a ratio is not a factor of safety, whatever it comes out at.

ASK 10W

Estimate the observed capacity-to-demand ratio for the two tested anchors.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a designed capacity is a claim to be checked

VERDICT 103W

Two anchors were pulled. They averaged 25.5 kN against a demand of 31 kN, so what was measured is about 0.82 of what the wall asks of them. The anchors are short. But 0.82 is not a factor of safety, and calling it one claims far more than two pulls can carry: a real check also has to hold the scatter between anchors, the way the joint lets go when it lets go, and the margins the code puts on both. None of that has to be settled to act. Restrict the gym, test enough anchors to mean something, and design the repair.

TAKEAWAY 26W

Two pull tests can show a serious capacity shortfall; they do not by themselves establish a code-level factor of safety for every anchor in the building.

END OF THE DAY 18W

Ranking is about which problem grows fastest and which is cheapest to remove, not about which sounds worst.

The last morning it can be changed

PLAN CARD 121W

Thursday, and the recovery plan goes to the council tonight. Inside it is one clause that matters more than the rest: whether rebuilding in the Flats requires ground improvement, or merely stronger foundations. Elena Navarro has the corrected amplification and the borehole programme. Bram Halvorsen has the cost, which is large and falls on people who have just lost houses. Yvette Delacroix has 400 households who want to rebuild this year and cannot afford to wait for a scheme. After tonight the clause cannot be reopened for five years. Today you work out what ground improvement buys, what it costs, and what the town commits to if the clause goes in — and what it commits to if it does not.

BRIEFING 16W

The recovery plan closes for comment today, and the ground improvement clause is in or out.

WHAT YOU DECIDE 12W

Commit the town to a rebuilding standard while the window is open.

14.1 Fixing the ground instead of the building

GEOTECHNICAL · A ROOM · CHOICE

SCENE 33W

Navarro has three options costed: stone columns, deep densification, and doing nothing to the ground while designing stiffer foundations on top of it. The council wants one approach written into the rebuild guidance.

GUIDE 69W

Three costed options and four readings of what separates them. Ask of each option where in the system it acts: on the soil, on the structure, or on neither. A stiff foundation can carry a building across ground that moves. It does nothing for the road, the buried services or the site next door. Both approaches can be right, and the council is writing one of them into guidance.

ASK 16W

What is the engineering difference between treating the ground and designing the foundation around the hazard?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

liquefaction can be addressed in the ground or accommodated in the structure

VERDICT 88W

The two strategies act at different places in the system. Densification can increase density and cyclic resistance; drains or stone columns may also help dissipate excess pore pressure, depending on the design. That reduces liquefaction susceptibility rather than guaranteeing the soil can never liquefy. Structural solutions such as piles or rafts may bypass weak layers or tolerate differential movement, but they leave roads, buried utilities and neighbouring sites exposed to ground deformation. Both

can be legitimate; the town-level question is which consequences the policy is trying to reduce.

TAKEAWAY 18W

Ground improvement reduces liquefaction susceptibility or its consequences; structural foundations can instead bypass or accommodate some ground deformation.

14.2 A measurable acceptance target

MATERIALS & TESTING · A PERSON · BALLPARK

SCENE 35W

Ferreira has cone penetration results from a trial panel: the fill tested at 4 MPa before the stone columns went in and 11 after. The two CPT traces lie side by side on the desk.

GUIDE 58W

Five numbers, and three belong elsewhere: the depth of the fill, the amplification factor and a concrete strength. Ask of each whether it came from these two cone traces. And note what the factor is and is not. Cone resistance is an input to a liquefaction assessment, so 2.75 times the resistance is not 2.75 times the margin.

ASK 9W

By what factor did the measured cone resistance increase?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

CPT stands for cone penetration test: a standard cone is pushed into the ground and its resistance is recorded

VERDICT 84W

The measured cone resistance rose from 4 to 11 MPa, a factor of 2.75. That is a strong, auditable construction-control result: the treated ground is substantially more resistant to penetration at that location. But q_c is an input to liquefaction evaluation, not a direct safety factor. Cyclic resistance also depends on overburden, fines, groundwater and the design method used. A good specification can therefore require a target corrected CPT resistance or verified performance profile without claiming that '2.75 times q_c ' means '2.75 times safer.'

TAKEAWAY 21W

CPT resistance is a measurable quality-control indicator; a 2.75-fold increase in q_c is not automatically a 2.75-fold increase in liquefaction resistance.

14.3 What the next five years hold

PUBLIC SAFETY · A ROOM · PROTOCOL

SCENE 35W

The clause goes in or it does not, and nobody can reopen it for five years. Delacroix is holding the spring rebuild schedule beside the draft ordinance. The geotechnical section is the last unresolved page.

GUIDE 63W

Four versions of the clause on the left, and what each one actually produces on the right. Pair them by asking who has to pay and how soon. A requirement nobody can afford stops rebuilding instead of improving ground. No requirement rebuilds the same vulnerability on the same fill. And the last pairing is about time: nobody can reopen this for five years.

ASK 7W

Commit, and say what the commitment contains

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a decision to require something is also a decision about who pays for it

VERDICT 85W

The engineering evidence can define the ground hazard and the performance a treatment programme should demonstrate. It cannot decide who pays or how quickly households must comply. A requirement without financing can stall rebuilding. No requirement can reproduce the same vulnerability. A workable policy therefore states where the rule applies and names a measurable geotechnical acceptance target. It also provides funding or phasing and a review process as new site data arrive. That separates the technical standard from the political choices needed to implement it.

TAKEAWAY 21W

Science can define the hazard and performance target; the rebuilding policy must also specify affordability, phasing and who carries the cost.

14.4 What eight degrees does

STRUCTURAL ASSESSMENT · A ROOM · CALLBACK · CHOICE

SCENE 38W

Okonkwo has the column loads. At eight degrees the weight above each column no longer lands over its centre, and the offset grows with height up the building. The survey sheet and column plan are open beside him.

GUIDE 65W

All four options are true things about a leaning building. They differ in whether they name a load path or a feeling. Ask of each whether an engineer could put a number on it from the survey sheet and the column plan. A column carries weight best along its own axis. Eight degrees moves that weight off the axis, and the offset grows with height.

ASK 10W

What is the engineering objection to leaving Marina Court standing?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a leaning column carries its load off-centre

VERDICT 81W

A column is efficient when its gravity load stays close to its axis. Once the building leans, the weight above acts with an offset, creating additional bending moment; as lateral displacement grows, the gravity load can amplify that demand through P- Δ effects. The engineering question is therefore not whether eight degrees looks frightening, but whether the measured geometry, member capacities and continued ground stability leave adequate margin. The undamaged frame is encouraging evidence, not permission to ignore the new load path.

TAKEAWAY 19W

The lean matters because gravity now acts with eccentricity, creating $P-\Delta$ effects and extra bending demand in the frame.

END OF THE DAY 19W

The moment a decision stops being reversible is the moment it needs the whole argument, not the quickest one.

What we know, and how well

BEFORE THE DAY — TAG

Catch Ives before the hospital handover

Ives, the hospital facilities manager, has the only copy of the ward occupancy plan with the yellow-placard rooms marked on it, and he is walking back to the hospital for the ten o'clock handover. After that he is on a ward round for four hours, and the afternoon's decision about the ground floor rests on a drawing that currently exists once.

PLAN CARD 119W

Friday, two weeks on, and the fortnight goes into a report. Marisol Okonkwo will not sign a document that states everything at the same confidence, because a reader who cannot tell a measurement from a judgement will act on both the same way. Elena Navarro has the borehole results. Rei Tanaka wants her forecast stated as a rate with a range rather than as a number. Inês Cardoso wants the vault siting error written down as a finding rather than buried as a correction. Funmi Adeyemi has to turn all of it into something a household can use. Today you sort the findings by the evidence under them, and decide which of the fortnight's changes should outlive the emergency.

BRIEFING 20W

The report goes out today, and it has to say which of its findings are measurements and which are judgements.

WHAT YOU DECIDE 11W

State the fortnight's findings at the confidence each one has earned.

15.1 Measured, inferred, assumed

SEISMIC NETWORK · A ROOM · CASEBOOK

SCENE 32W

Cardoso lays out four of the fortnight's statements. The office believes all four. They are not held for the same reasons and she will not let them into the report as equals.

GUIDE 63W

Four statements the office believes, held for four different reasons. Sort them by what an opponent would have to produce to challenge each: a second log, another event, a borehole, or a future earthquake. Belief is not the variable. The weakest of the four is the one the plan clause rests on, and saying so in the report is what makes it usable.

ASK 6W

Sort the findings by their evidence

BEHIND THE BUTTON 181W

Why explanations rather than labels. Naming a finding is not accounting for it, and a plausible-sounding mechanism attached to the wrong observation is the commonest way a wrong story survives. Committing an explanation to one clue means claiming it accounts for that clue specifically and not for its neighbour, which is where the two come apart.

How to use the one-each rule. The explanations are a set to be distributed, not a list to be sampled, so every join constrains the rest. Settling the two you are confident of can decide the remaining pair by elimination. Where it does not, two explanations are still competing for one clue, and that competition is the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every explanation used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a claim can rest on a measurement, on an inference, or on nothing yet

VERDICT 86W

The four claims deserve different labels because their evidence chains differ. The vault siting error is directly documented and independently checked against competent rock. The hospital's 0.31 g is a direct measurement for 1 location and 1 event. Marina Court's mechanism is a strong inference from survey, ejecta and geotechnical behaviour rather than a direct observation beneath the raft. Future Flats amplification is conditional on corrected site-response evidence and the frequency content of a future earthquake. Stating those limits makes the report more useful, not weaker.

TAKEAWAY 17W

Sorting findings by their evidence is what lets a later reader know which ones to check first.

15.2 The number that has to be a range

HAZARD & FORECASTING · INCIDENT BASE · A PERSON · CHOICE

SCENE 35W

Tanaka's aftershock forecast goes into the report. She wants the range in the sentence and Okonkwo wants to know what the range is for. The staffing appendix cannot be signed until she chooses the wording.

GUIDE 68W

Four reasons to publish a range, and they differ in who the range is for. Ask of each whether it is about caution, honesty, defensibility, or planning. Nobody staffs an emergency to the average week. A single number tells a planner nothing about how far the sequence might run above it, so they invent a margin instead. The staffing appendix cannot be signed until the wording is chosen.

ASK 14W

Why does the forecast go in as a range rather than a single number?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a forecast is a distribution rather than a single value

VERDICT 82W

Nobody staffs an emergency to the average week. They staff it to a week that is worse than average, and how much worse is exactly what the range says. A single number tells a planner nothing about how far the sequence might run above it, so the planner invents a margin — usually badly, usually too small. The range is also the honest part: it says which part of the behaviour is well constrained by 400 recorded events and which is not.

TAKEAWAY 17W

A forecast without a range cannot size a decision, because the decision is sized against the range.

15.3 What outlives the emergency

STRUCTURAL ASSESSMENT · A ROOM · TRIAGE

SCENE 35W

Four things the office started doing under pressure. Okonkwo asks which of them is worth its cost on an ordinary Tuesday, when nothing has happened. The emergency procedures are being rewritten into the permanent manual.

GUIDE 62W

All four practices are good ones. Two questions separate them. Does it block the failure that actually happened? And will the office still be doing it in a quiet month? Trace each back to the mechanism. The gym, the plant room and the basement were all outside the first inspection. All three omissions were written down, and nobody acted on the list.

ASK 17W

Which permanent practice would most directly prevent the school, hospital and Ferry Street failure mode from repeating?

BEHIND THE BUTTON 194W

Why first is a different question from most important. When several calls compete for the same hour, what a choice is worth is not its own importance but the difference between doing it now and doing it later. A serious problem that will be no worse in an hour costs nothing to defer. A smaller one that closes a door costs everything behind that door.

What to look for in the options. Two things separate them: which is still changing, and which is a precondition for the others. A situation that is deteriorating has a cost per hour attached to it, and a step that unblocks the rest multiplies the value of the hours after it. Everything else is a preference about where to start.

Why only one answer is marked. In the situation these options describe all of them eventually happen; what is being tested is the head of the queue, because that is where the reasoning is visible. The verdict names what each of the others was waiting on, which is worth reading even when the choice was right — the ordering behind the first place is the rest of the answer.

TAKES AS READ

a practice adopted under pressure has a cost that only shows later

VERDICT 90W

A permanent practice should catch recurring failure modes at a cost the office will actually keep paying. The school gym, hospital plant room and Ferry Street basement were all outside the first inspection scope, yet the omissions were written down. Making that unresolved list actionable would have exposed each gap. Two engineers on every placard would halve throughput without a measured benefit. Daily public meetings were useful only during the crisis. Re-inspecting every shored building after every magnitude-4 event is so broad that staff would eventually stop following the rule.

TAKEAWAY 23W

An unknown only becomes safer when it has an owner, a deadline and a rule that prevents final clearance from silently forgetting it.

15.4 What knocking it down would fix

PUBLIC SAFETY · A ROOM · CALLBACK · VALUE

SCENE 36W

Halvorsen wants Marina Court demolished this week. Delacroix has 12 neighbouring buildings on the same reclaimed ground and 5 investigation-days before the cordon review. Four proposed uses of that window are written across the site map.

GUIDE 61W

Five investigation-days, four proposals, and a demolition somebody wants this week. Open each proposal to see what decision its result could change. Marina Court is already cordoned, so evidence about that one building changes nothing that is not already decided — the twelve neighbours on the same reclaimed ground are where the open decision is. Commit the window you would spend.

ASK 12W

Which investigation buys information that generalises beyond the one building already cordoned?

BEHIND THE BUTTON 148W

What is already decided and what is not. Marina Court is cordoned; nobody is going back in this week whatever a test says. The open question is whether the ground condition that damaged it requires action on the twelve buildings around it, and that is a question about the ground rather than about the building.

Why the most visible damage is the trap. A proposal to instrument the worst-damaged structure produces vivid results and answers a question that has been closed. It is the most defensible-sounding way to spend five days and learn nothing that changes a decision.

What generalising means here. A campaign of cone tests and boreholes across the Flats measures the fill itself, which every one of the thirteen buildings sits on. One result then bears on twelve open decisions instead of one closed one — the same money, an order of magnitude more use.

TAKES AS READ

a decision can treat a symptom or a cause

VERDICT 81W

Demolition can remove Marina Court as a visible hazard, but it does not reveal whether the same ground condition extends beneath nearby buildings. A corridor investigation can. CPTs, boreholes and groundwater checks test the ground shared by several sites while the existing cordon controls the known building hazard. Extra lean monitoring or a one-building underpinning concept answer narrower questions. Value of information therefore depends on the decision the evidence can change, not on how dramatic or precise the proposed work sounds.

TAKEAWAY 26W

The best next measurement is the one that can change the decision you still have to make, not the one that studies the most visible damage.

END OF THE DAY 21W

A finding is worth what its evidence is worth, and saying so is what makes a report usable a year later.

THE CLOSING CARD

The last cordon came down in the ninth week, street by street, each one when the hazard written against it had been dealt with rather than on a date anybody promised. The hospital never closed. Bay Road School lost its gymnasium for a term and kept its classrooms. Every parapet on the Parade is tied back now, which was done while the scaffolding was still up and cost a third of what it would have a year later.

What it cost: two weeks of a town living in halls, a six-storey block that will be taken down in the spring, and a recovery plan whose ground-improvement clause four hundred households have to find the money for. What is unfinished: the vault on the bench is still the reference station and still on weathered granite, the corrected amplification is a projection from nine boreholes, and the practice that would have caught all of it — writing down what an inspection could not see, and acting on that list — is a paragraph in a report until the next coordinator makes it a habit.

Card text only — no options, boards or answers. 13,660 words of card prose across 61 stops. Generated from themes/aftershock by tools/make-cardbook.mjs.